

NORTHPOINT

# Northpoint Digital Innovation Fund

---

Six-year performance review · January 2017 – December 2022

Final report to investors · New York · San Francisco · London · Hong Kong ·  
Shanghai · Sydney

Callum O'Connor · Portfolio Manager

Student ID 14053836 · FINC13-303 Portfolio Analysis and Investments · Bond University

MANDATE OUTCOME

## \$279m

from a \$100 million mandate

+179.2% cumulative  
18.7% a year

6.6% alpha,  $t = 3.06$

# Agenda

Six sections, twenty-five minutes, then questions

- |   |                              |                                                                              |
|---|------------------------------|------------------------------------------------------------------------------|
| 1 | Where we started             | The mandate, the theme, and how the fund was actually run                    |
| 2 | The economy we lived through | Six years, three regimes, and how our 2016 scenarios held up                 |
| 3 | What the fund returned       | Wealth created, benchmarked, and stress-tested through COVID and 2022        |
| 4 | How much risk we took        | Distribution, risk-adjusted returns, tail risk, and seasonality              |
| 5 | Was it skill?                | Jensen's alpha, factor models, attribution, and the optimised counterfactual |
| 6 | The verdict                  | Objective scorecard, fees, and our recommendation on the fund's future       |

# The six-year result

Every figure on this slide is calculated from 72 months of dividend-adjusted total returns and is reproduced in the appendix

---

## +179.2%

### CUMULATIVE RETURN

\$100m mandate grew to \$279.2m.  
S&P 500 returned +90.0%;  
NASDAQ-100 +134.7%.

---

## 18.7%

### ANNUALISED RETURN

Against a target of 8-10% net p.a.  
S&P 500 11.3%; NASDAQ-100 15.3%.

---

## 0.94

### SHARPE RATIO

Highest of every comparator.  
NASDAQ-100 0.75; S&P 500 0.65.  
Sortino 1.51.

---

## -24.9%

### MAXIMUM DRAWDOWN

Shallower than our own benchmark.  
NASDAQ-100 fell 32.6%.

---

## 6.6%

### ALPHA P.A. (FF5)

Fama-French five-factor Jensen's  
alpha,  $t = 3.06$ ,  $p = 0.002$ .  
Statistically significant.

---

## 1.04

### BETA TO THE MARKET

Market-like sensitivity.  
The excess return is not  
simply leverage on the index.

---

## 0.98

### INFORMATION RATIO

Active return per unit of  
tracking error vs the S&P 500  
(6.98% tracking error).

---

## 50 / 72

### POSITIVE MONTHS

A 69.4% hit rate.  
Average up month +4.6%,  
average down month -5.2%.

# 01

---

## Where we started

The mandate you funded in 2016, the eleven positions we bought, and whether we stayed faithful to the brief

# The mandate you funded

Nothing in the strategy changed over six years. The discipline was in the rules, not in the reacting

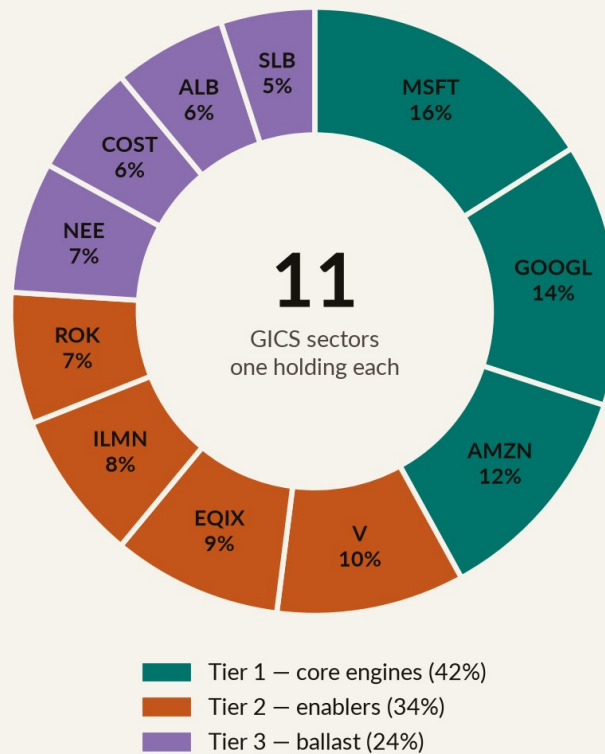
| MANDATE PARAMETER   | WHAT WE COMMITTED TO IN JANUARY 2016                                              |
|---------------------|-----------------------------------------------------------------------------------|
| Capital             | US\$100 million, six-year fund life (Jan 2017 – Dec 2022)                         |
| Strategy            | Active, high-conviction thematic equity — technology and innovation               |
| Theme               | Digital transformation: cloud, mobile data, payments, automation, genomics        |
| Holdings            | Eleven US-listed stocks — exactly one per GICS sector                             |
| Weighting           | Conviction tiers, 5% floor per sector, 16% cap on the largest position            |
| Rebalancing         | Quarterly, plus a 5-percentage-point drift threshold; mid-life review at end-2019 |
| Primary benchmark   | NASDAQ-100 (theme-aligned — measures selection skill within the theme)            |
| Secondary benchmark | S&P 500 (broad market — measures whether the theme was worth owning)              |
| Fees                | 0.90% p.a. management + 10% of returns over the hurdle, high-water mark           |
| Return target       | 8–10% net p.a., roughly 55–70% cumulative net over the fund life                  |

## WHERE WE STARTED

# Eleven sectors, one idea

The theme was expressed through conviction tiers, never by abandoning a sector

Target weights by conviction tier



## HOW THE BOOK WAS BUILT

**Tier 1, 42%.** Microsoft, Alphabet and Amazon: the platform companies where the cloud, advertising and e-commerce theses all compound at once. Highest conviction, largest weights.

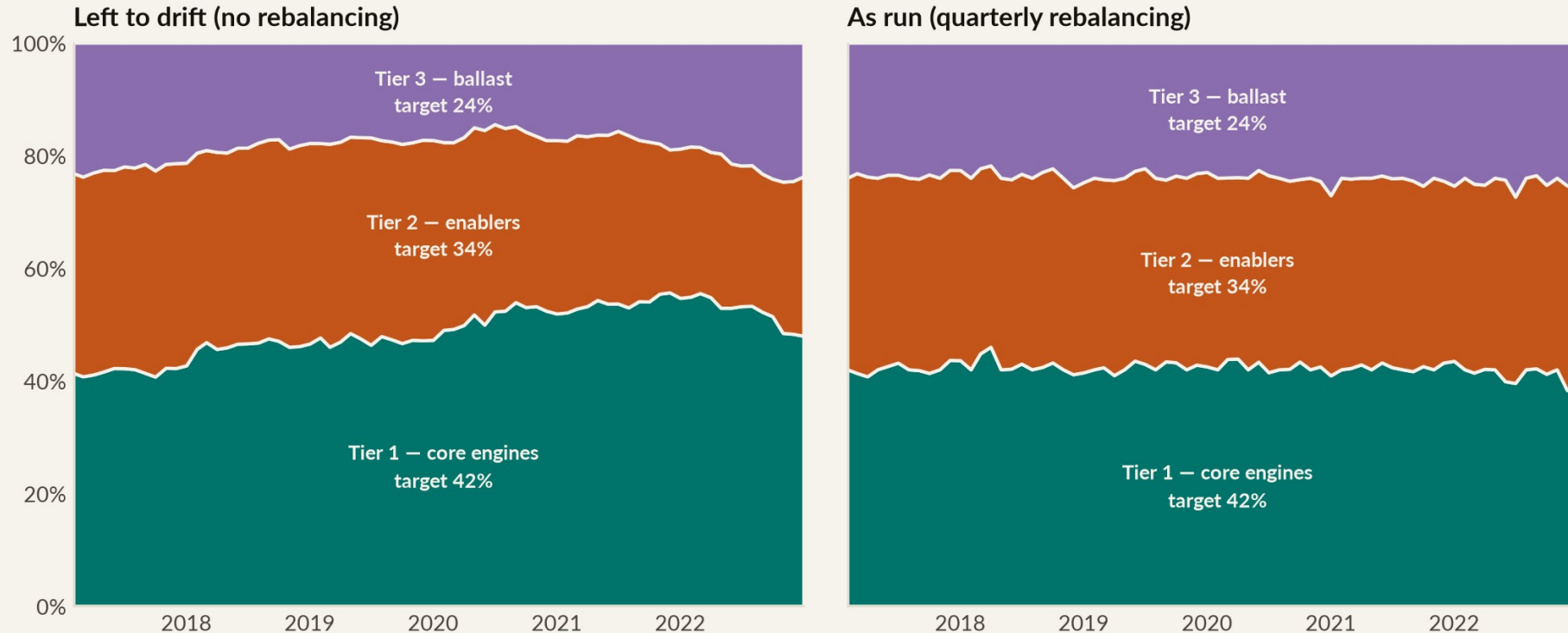
**Tier 2, 34%.** Visa, Equinix, Illumina and Rockwell: the enablers. Each monetises digitisation in one specific channel — payments, data centres, genomics, factory automation.

**Tier 3, 24%.** NextEra, Costco, Albemarle and Schlumberger: diversification and ballast, each still carrying an innovation or recovery angle.

**The floor mattered.** Energy at 5% and Materials at 6% looked like drag for five years. In 2022 they were the two reasons our drawdown ran ten points shallower than the NASDAQ's.

# We rebalanced. That was a decision, and it paid.

Quarterly rebalancing added 1.3 percentage points a year over letting the winners run



|                                       | CUMULATIVE | CAGR  | SHARPE | MAX DRAWDOWN | TERMINAL ON \$100M |
|---------------------------------------|------------|-------|--------|--------------|--------------------|
| <b>Quarterly rebalanced (as run)</b>  | 179.2%     | 18.7% | 0.94   | -24.9%       | \$279m             |
| <b>Left to drift (buy &amp; hold)</b> | 160.9%     | 17.3% | 0.88   | -27.8%       | \$261m             |

# 02

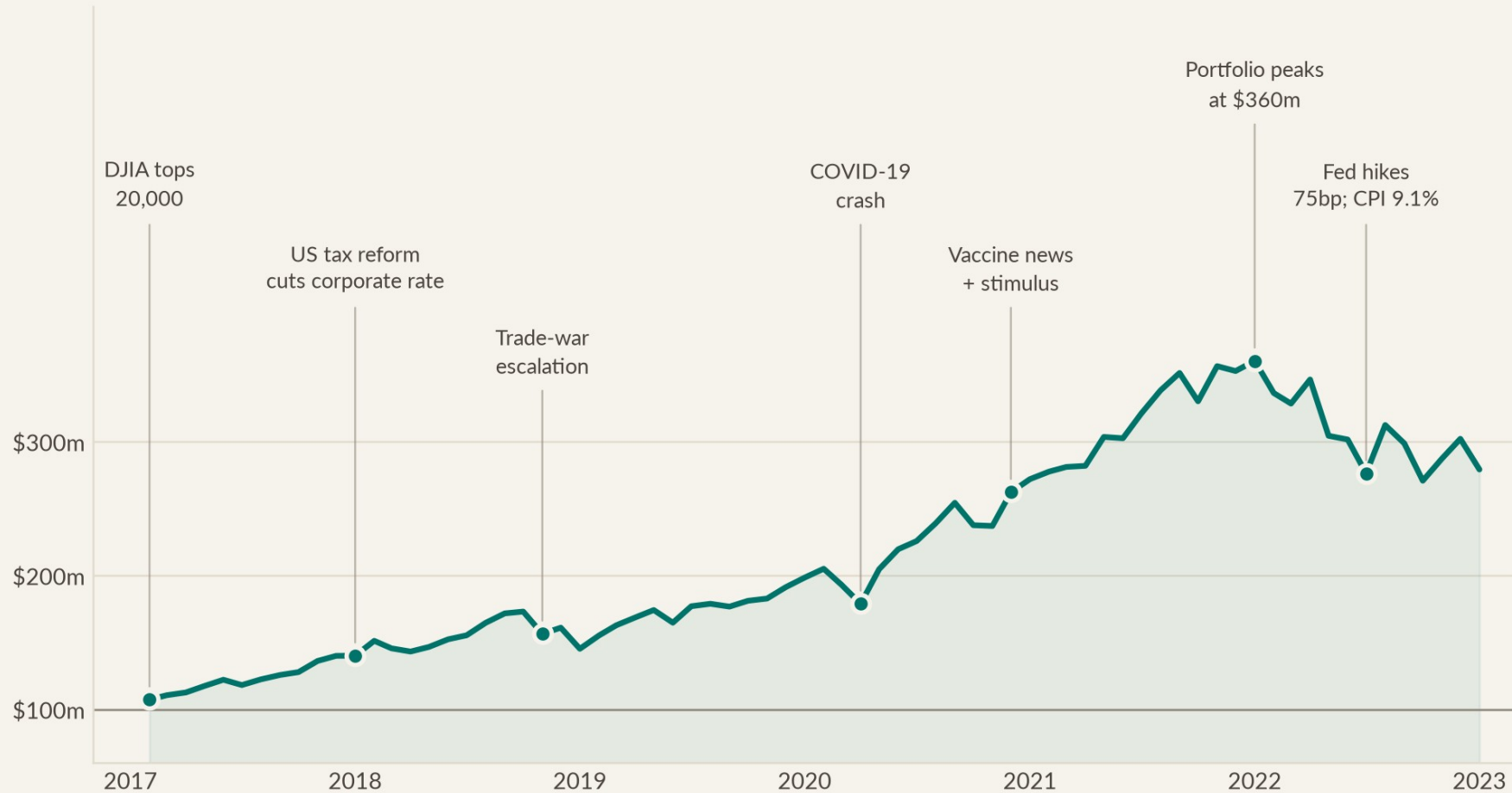
---

## The economy we lived through

Three distinct regimes in six years — and an honest mark-to-market of the scenarios we published in January 2016

# Six years, seven turning points

The events that shaped the fund, plotted against the wealth they created or destroyed



## THE ECONOMY WE LIVED THROUGH

# The macro backdrop, regime by regime

How the state of the economy translated into the portfolio's returns

## 2017 – 2019

### SYNCHRONISED GROWTH

**2017: +40.0%** Synchronised global growth, benign inflation, a gradual Fed. US tax reform cut the corporate rate from 35% to 21%, lifting after-tax earnings across the book.

**2018: +3.7%** Trade-war escalation and a hawkish Fed produced a violent Q4. We lost 9.6% in October and 9.8% in December — and still finished positive while the S&P fell 4.6%.

**2019: +36.6%** The Fed pivoted to cutting. Multiple expansion did most of the work; cloud and payments earnings kept compounding underneath it.

## 2020 – 2022

### PANDEMIC, STIMULUS, THEN THE RATE SHOCK

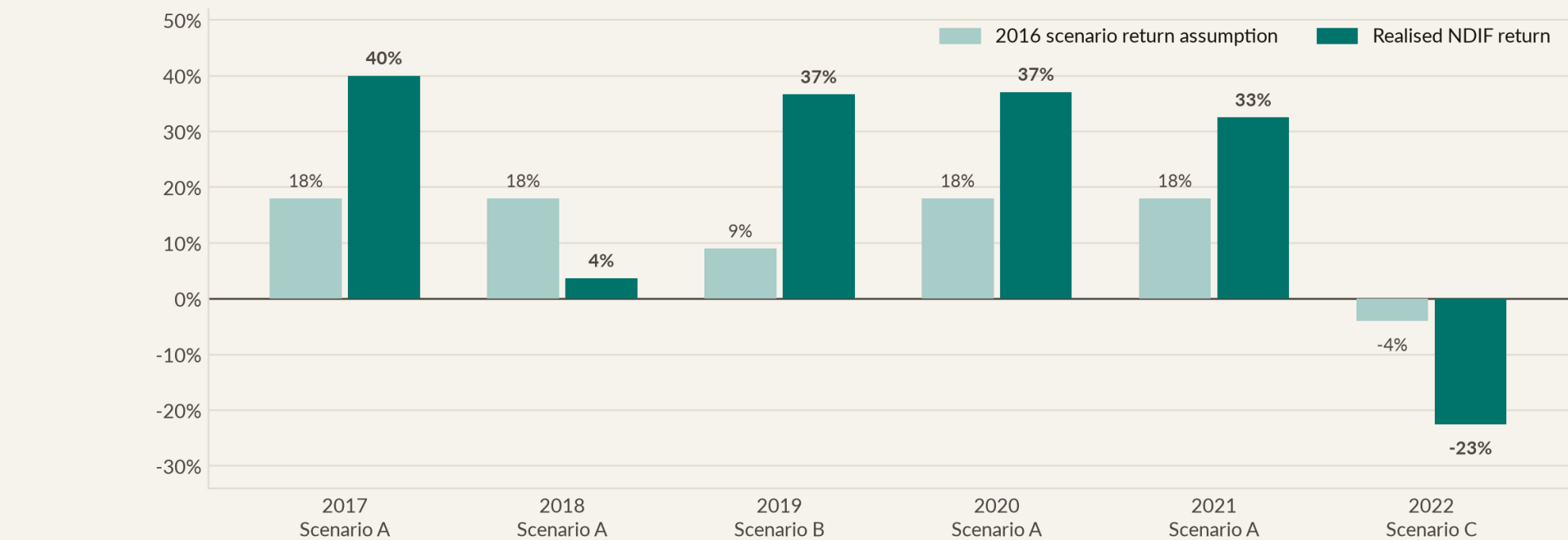
**2020: +37.1%** The COVID crash took 12.6% off the fund in two months. Then the pandemic became the single largest accelerant our thesis ever received: remote work, e-commerce and cloud migration pulled forward by years.

**2021: +32.5%** Reopening, record stimulus and peak growth valuations. Our thesis was consensus by now — which is usually the warning sign, and it was.

**2022: –22.5%** The sharpest tightening cycle since 1981. CPI peaked at 9.1%; the Fed raised 425bp. Long-duration growth equity de-rated hardest. Our energy and materials ballast is why we fell 10 points less than the NASDAQ-100.

# Marking our 2016 scenarios to market

We were right about direction in five of six years, and consistently too conservative about magnitude



| SCENARIO (AS PUBLISHED, JAN 2016) | ASSIGNED PROBABILITY | ASSUMED RETURN P.A. | YEARS IT ACTUALLY DESCRIBED | VERDICT                        |
|-----------------------------------|----------------------|---------------------|-----------------------------|--------------------------------|
| A — Growth & digital acceleration | 50% ST / 40% MT      | +18%                | 2017, 2018, 2020, 2021      | Occurred in 4 of 6 years       |
| B — Stagnation / low-growth grind | 30% ST / 30% MT      | +9%                 | 2019                        | Understated: we returned 36.6% |
| C — Recession / risk-off shock    | 20% ST / 30% MT      | −4%                 | 2022                        | Understated: we lost 22.5%     |

# 03

---

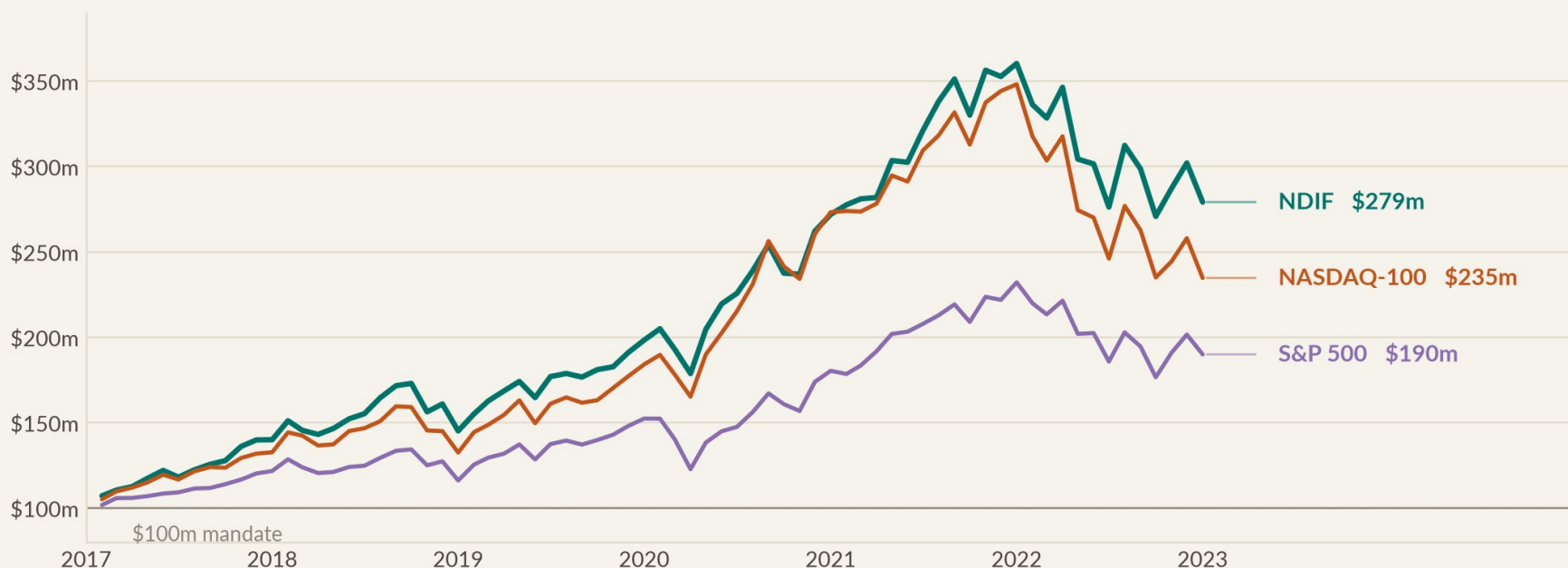
## What the fund returned

Wealth created, benchmarked against the indices you are paying us to beat, and stress-tested through the two events that mattered

## WHAT THE FUND RETURNED

## A \$100m mandate became \$279m

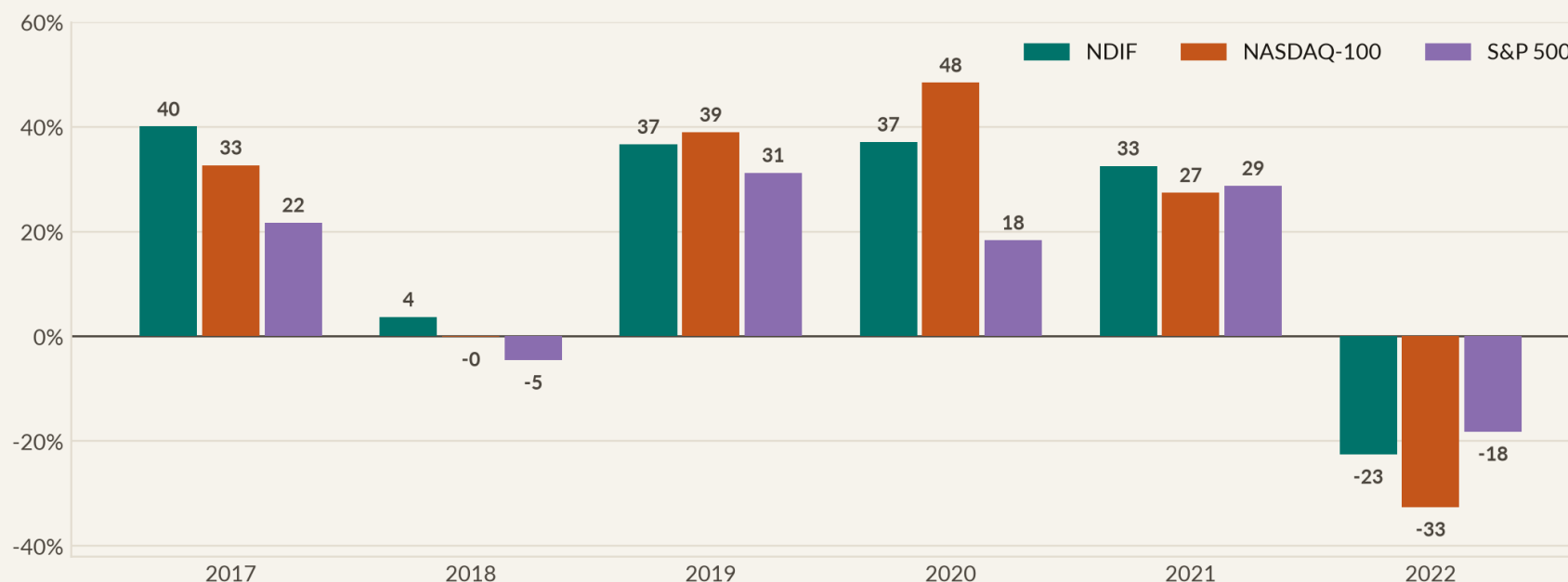
Ahead of the S&P 500 by 89 percentage points, and ahead of our own theme-aligned benchmark by 44



|                                           | CUMULATIVE | CAGR  | TERMINAL VALUE ON<br>\$100M | VALUE ADDED VS<br>NDIF |
|-------------------------------------------|------------|-------|-----------------------------|------------------------|
| <b>Northpoint Digital Innovation Fund</b> | 179.2%     | 18.7% | \$279.2m                    | —                      |
| <b>NASDAQ-100 (primary benchmark)</b>     | 134.7%     | 15.3% | \$234.7m                    | +\$44.4m               |
| <b>S&amp;P 500 (secondary benchmark)</b>  | 90.0%      | 11.3% | \$190.0m                    | +\$89.2m               |
| <b>60/40 stock/bond portfolio</b>         | 51.9%      | 7.2%  | \$151.9m                    | +\$127.3m              |

# Beat the S&P 500 in five of six years

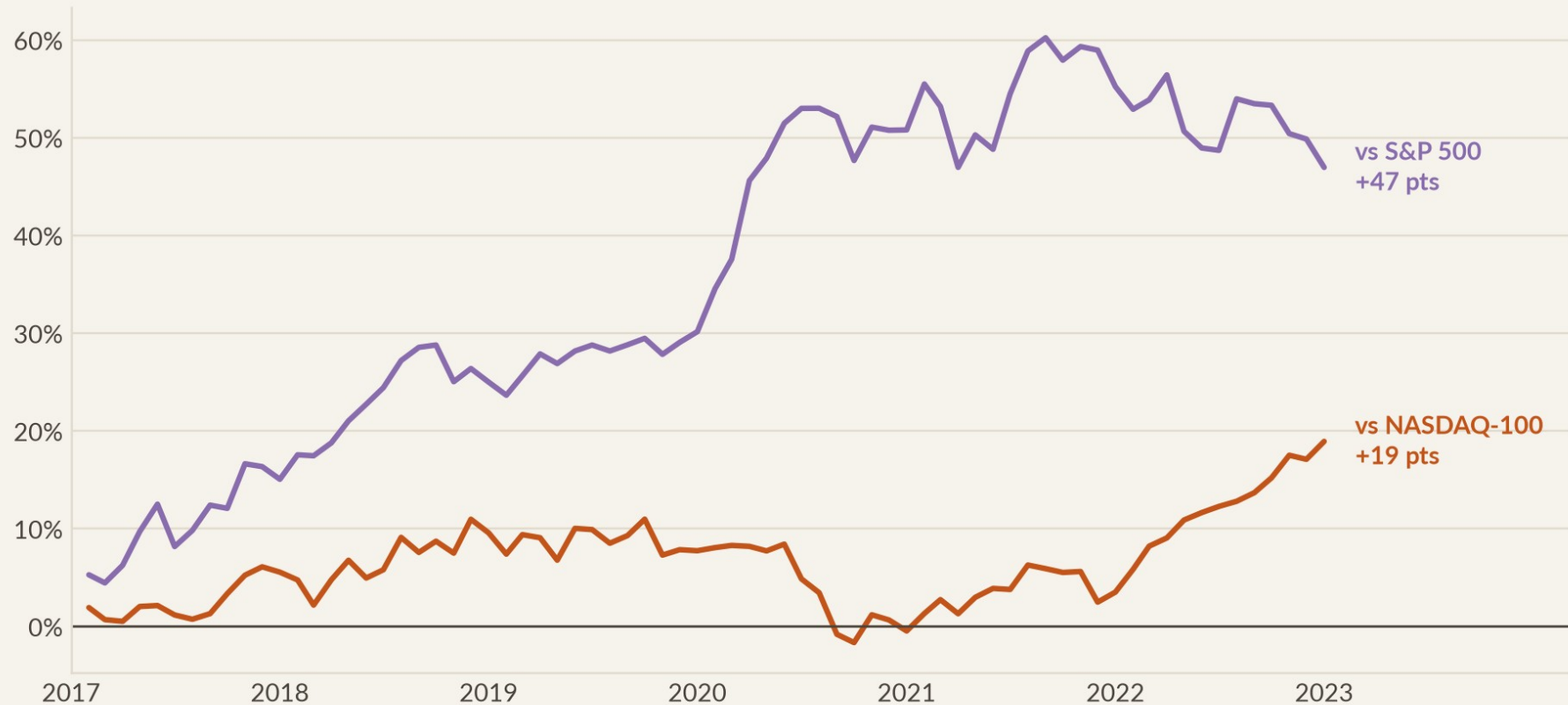
Including 2018 and 2022, the two years the market fell



| CALENDAR YEAR   | 2017   | 2018  | 2019  | 2020   | 2021  | 2022   |
|-----------------|--------|-------|-------|--------|-------|--------|
| NDIF            | 40.0%  | 3.7%  | 36.6% | 37.1%  | 32.5% | -22.5% |
| NASDAQ-100      | 32.7%  | -0.1% | 39.0% | 48.4%  | 27.4% | -32.6% |
| S&P 500         | 21.7%  | -4.6% | 31.2% | 18.3%  | 28.7% | -18.2% |
| NDIF vs S&P 500 | +18.3% | +8.2% | +5.4% | +18.8% | +3.8% | -4.4%  |

# The value-add compounded quietly, then paid off

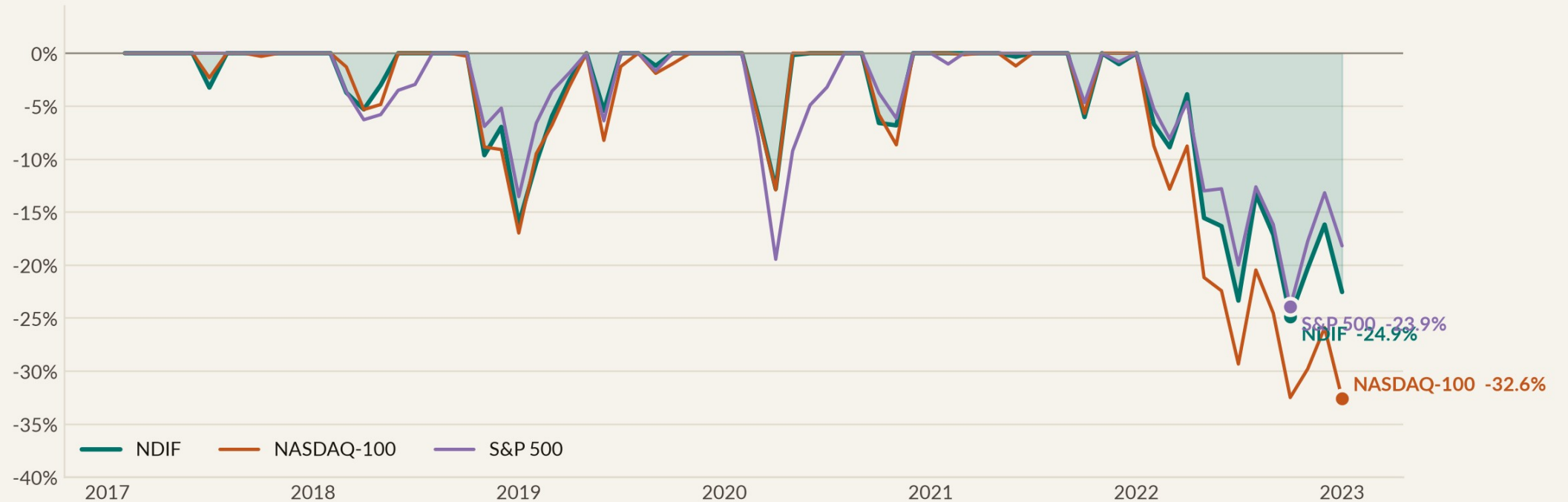
Cumulative performance relative to both benchmarks, rebased to zero at inception



Read this as the wealth ratio between the fund and each benchmark. A rising line means the fund is pulling ahead. Note that most of the outperformance versus the NASDAQ-100 was created after November 2021 — that is, on the way down, not on the way up.

# The fund fell less than its own benchmark

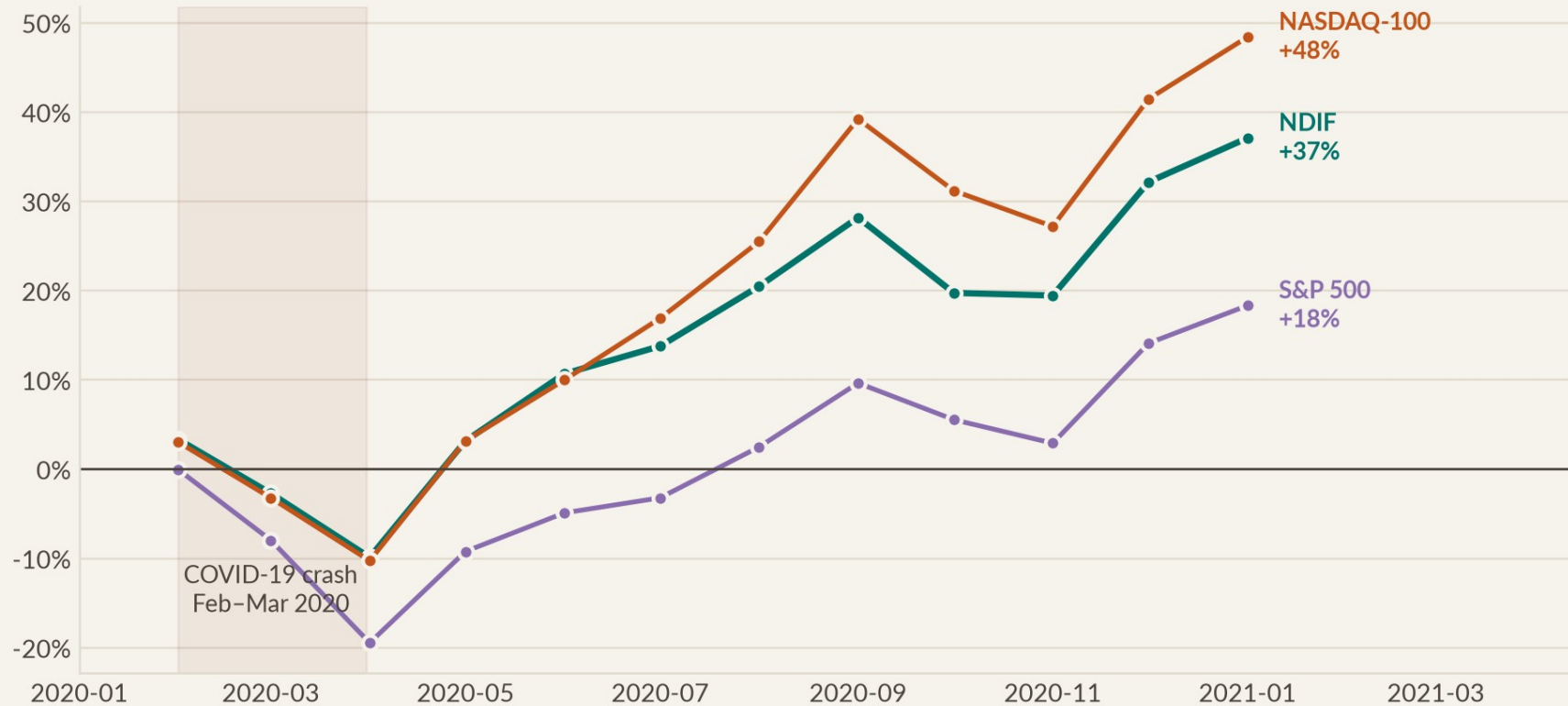
Maximum drawdown of –24.9% against the NASDAQ-100's –32.6%



|                    | MAXIMUM<br>DRAWDOWN | PEAK MONTH | TROUGH MONTH | CALMAR RATIO | DOWNSIDE<br>DEVIATION |
|--------------------|---------------------|------------|--------------|--------------|-----------------------|
| <b>NDIF</b>        | -24.9%              | Nov 2021   | Sep 2022     | 0.75         | 11.9%                 |
| <b>NASDAQ-100</b>  | -32.6%              | Nov 2021   | Dec 2022     | 0.47         | 12.9%                 |
| <b>S&amp;P 500</b> | -23.9%              | Dec 2021   | Sep 2022     | 0.47         | 11.3%                 |

# Stress test one: COVID-19

Scenario C arrived in March 2020. The fund lost 12.6% in two months and had recovered it within eight weeks

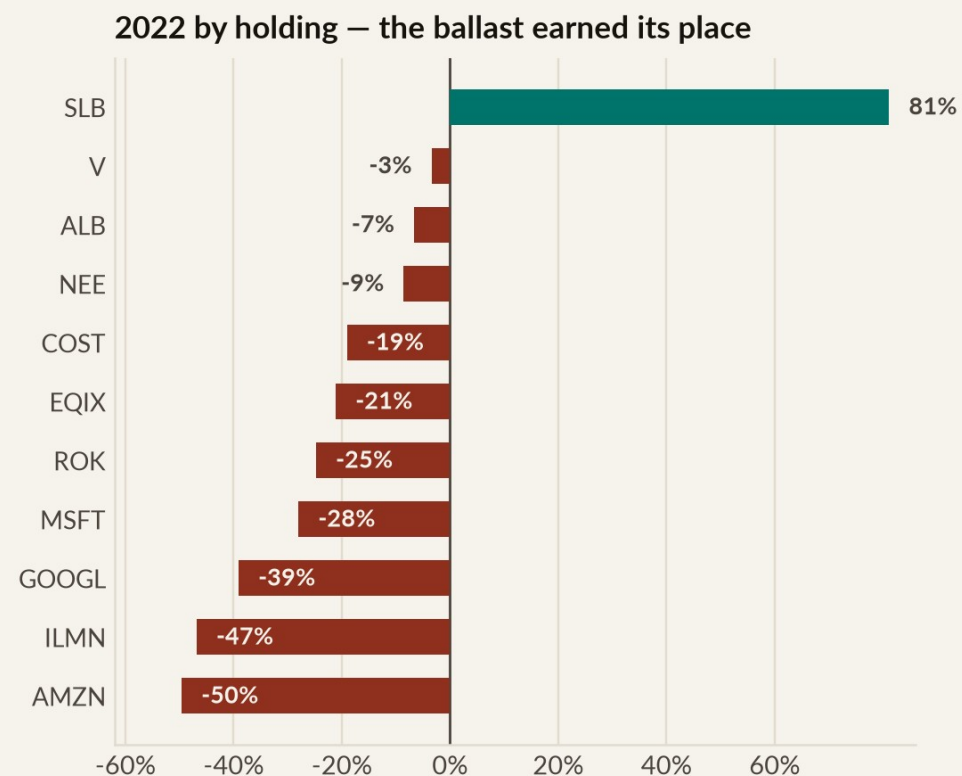
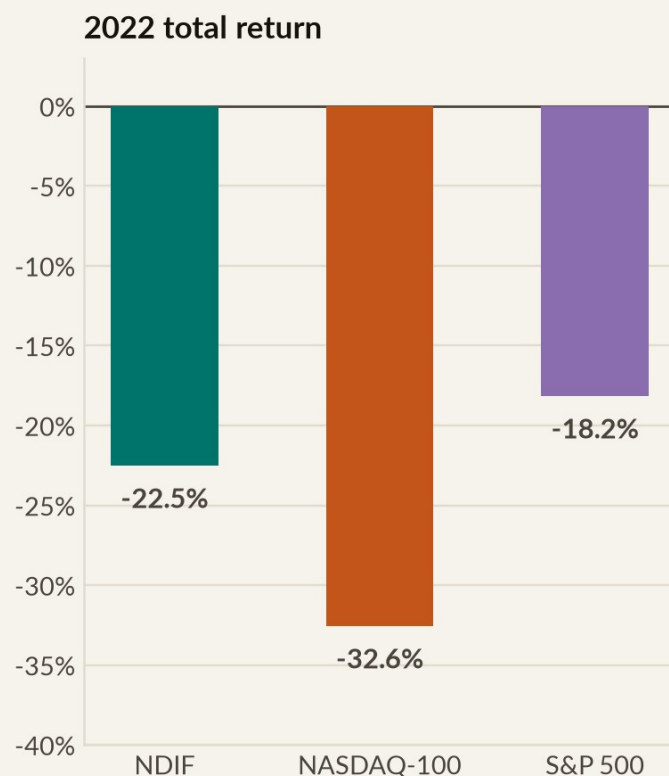


Our 2016 document had explicitly named “a systemic disruption such as a global pandemic” as a Scenario C trigger and assigned it a 20% probability. The thesis assumption that held up: cloud migration, e-commerce and payment digitisation are cost-saving, so they continue — and in fact accelerate — through a recession.

# Stress test two: the 2022 rate shock

Down 22.5% against a NASDAQ-100 down 32.6%, and the ballast is the entire reason why

What the fund  
returned



Schlumberger — the fund's worst holding over the full six years at  $-23.5\%$  — returned  $+81.2\%$  in 2022. Albemarle lost only 6.6% and NextEra 8.5%. Without the 5% energy floor and the 24% ballast tier, the fund's 2022 loss would have been close to the NASDAQ-100's. This is what a sector floor is for.

04

---

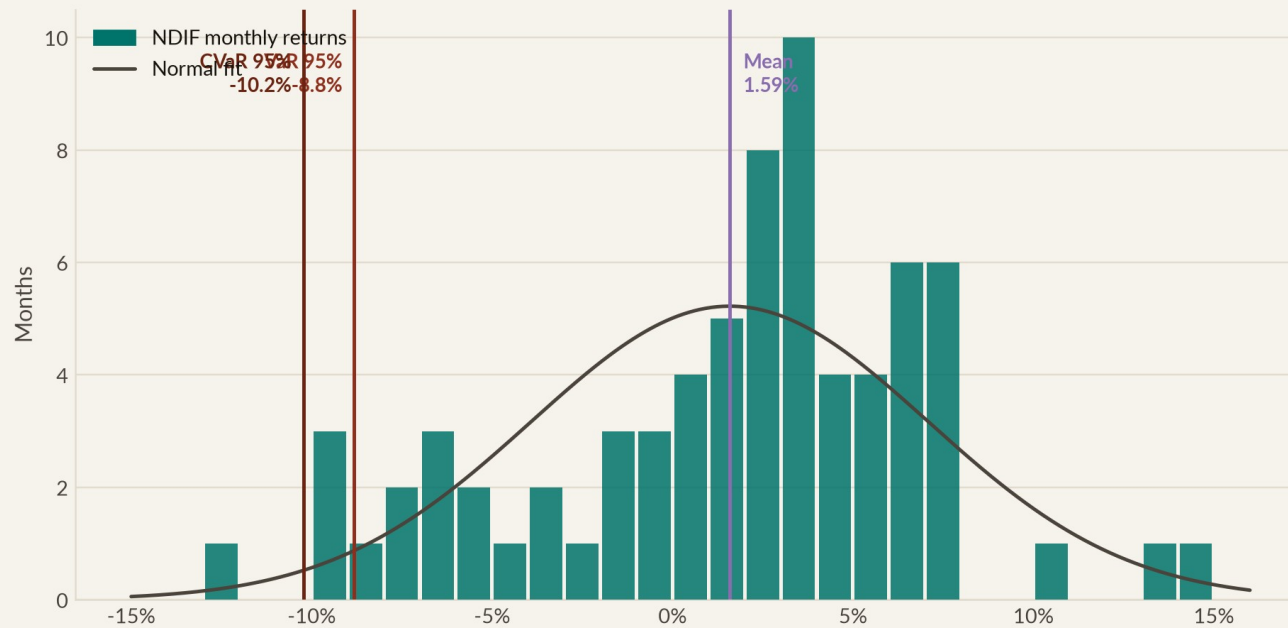
## How much risk we took

The full distribution of 72 monthly returns, the risk-adjusted measures, tail risk, and where in the calendar the money was made

## HOW MUCH RISK WE TOOK

# The shape of our returns

Close to normal, with a mild left tail. 72 monthly observations, Jan 2017 to Dec 2022



## DISTRIBUTIONAL STATISTICS

|                              |                         |
|------------------------------|-------------------------|
| Mean monthly return          | <b>1.59%</b>            |
| Median monthly return        | <b>2.73%</b>            |
| Standard deviation (monthly) | <b>5.50%</b>            |
| Annualised volatility        | <b>19.06%</b>           |
| Skewness                     | <b>-0.472</b>           |
| Excess kurtosis              | <b>0.067</b>            |
| Jarque-Bera p-value          | <b>0.278</b>            |
| Best month (Apr 2020)        | <b>14.5%</b>            |
| Worst month (Apr 2022)       | <b>-12.1%</b>           |
| Positive months              | <b>50 of 72 (69.4%)</b> |

## HOW MUCH RISK WE TOOK

# Risk-adjusted, we beat every comparator

Highest Sharpe, Sortino, Calmar and Treynor, and the highest information ratio

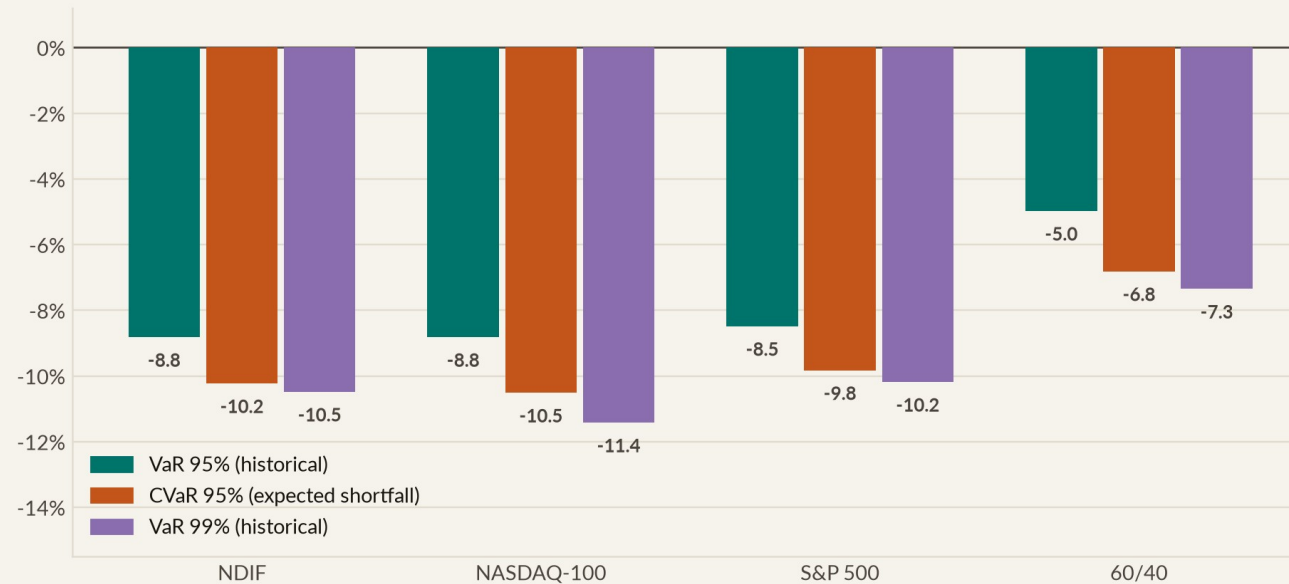
| MEASURE                                    | NDIF  | NASDAQ-100 | S&P 500 | EQUAL WEIGHT | 60/40 | WHAT IT TELLS YOU                        |
|--------------------------------------------|-------|------------|---------|--------------|-------|------------------------------------------|
| <b>Annualised return</b>                   | 18.7% | 15.3%      | 11.3%   | 18.0%        | 7.2%  | Geometric growth rate                    |
| <b>Annualised volatility</b>               | 19.1% | 20.2%      | 17.1%   | 19.4%        | 11.1% | Total risk                               |
| <b>Sharpe ratio</b>                        | 0.94  | 0.75       | 0.65    | 0.90         | 0.59  | Excess return per unit of total risk     |
| <b>Sortino ratio</b>                       | 1.51  | 1.18       | 0.98    | 1.45         | 0.88  | Excess return per unit of downside risk  |
| <b>Treynor ratio</b>                       | 0.173 | 0.139      | 0.111   | 0.163        | 0.102 | Excess return per unit of market beta    |
| <b>Calmar ratio</b>                        | 0.75  | 0.47       | 0.47    | 0.85         | 0.36  | Return per unit of maximum drawdown      |
| <b>Information ratio (vs S&amp;P)</b>      | 0.98  | 0.54       | —       | 0.97         | -0.72 | Active return per unit of tracking error |
| <b>Beta (vs S&amp;P 500)</b>               | 1.04  | 1.10       | 1.00    | 1.07         | 0.64  | Market sensitivity                       |
| <b>M<sup>2</sup> (risk-matched return)</b> | 17.2% | 14.0%      | 12.2%   | 16.5%        | 11.1% | Return at benchmark risk                 |

Risk-free rate is the Fama–French one-month Treasury bill series (mean 1.25% p.a. over the period). All ratios are computed on the same 72 monthly observations and annualised by  $\sqrt{12}$ .

## HOW MUCH RISK WE TOOK

# Tail risk sits below the NASDAQ-100 on every measure

One-month value-at-risk and expected shortfall from the empirical distribution

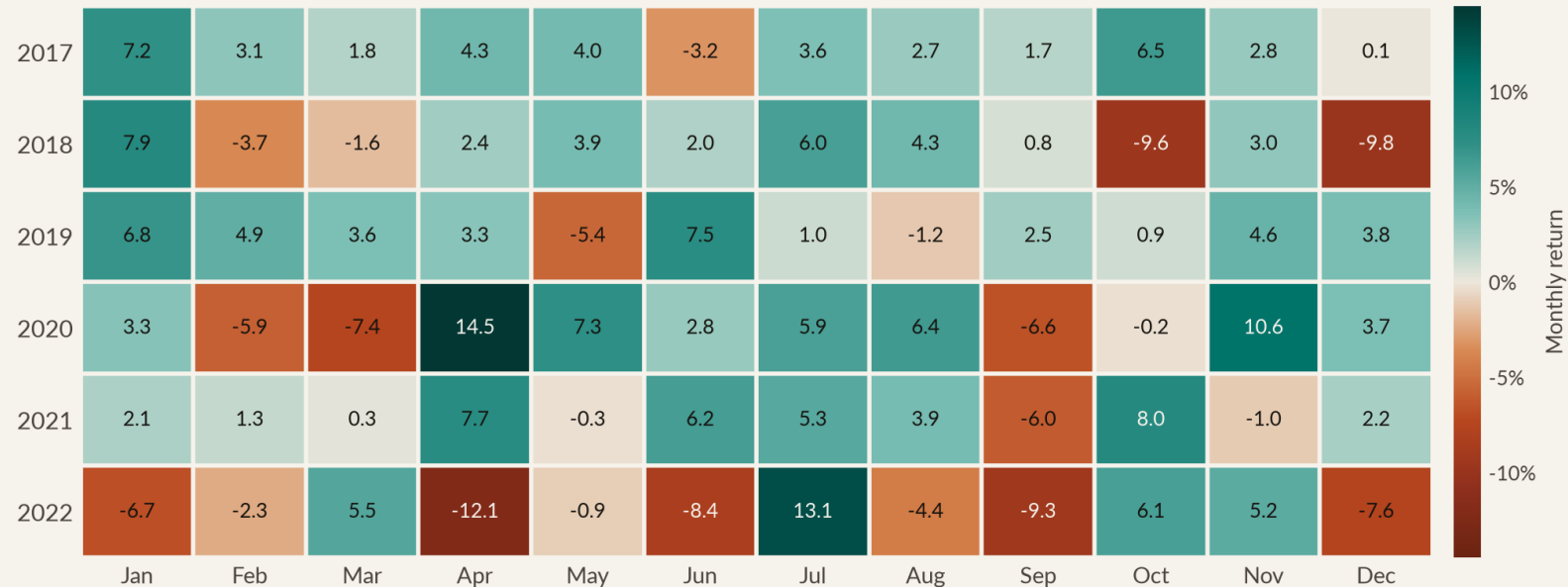


## NDIF TAIL MEASURES

|                               |                 |
|-------------------------------|-----------------|
| VaR 95% (historical)          | <b>-8.82%</b>   |
| VaR 95% (parametric normal)   | <b>-7.46%</b>   |
| VaR 95% (Cornish-Fisher)      | <b>-8.17%</b>   |
| CVaR 95% (expected shortfall) | <b>-10.23%</b>  |
| VaR 99% (historical)          | <b>-10.48%</b>  |
| CVaR 99% (expected shortfall) | <b>-12.14%</b>  |
| On \$279m, a 95% VaR month is | <b>-\$24.6m</b> |

# Where in the calendar the money was made

Seasonality across 72 months. Six observations per calendar month, so read this as description rather than a tradeable signal



## July +5.8%

The strongest month, on the back of six positive Julys out of six — including +13.1% in July 2022 during the bear-market rally.

## September -2.8%

The weakest month. Negative in three of six years, but heavily so: September 2022 alone cost 9.3%.

## The extremes cluster

April 2020 was the best month on record at +14.5%; April 2022 the worst at -12.1%. Both sit inside the same two-year window.

# 05

---

## Was it skill?

Jensen's alpha, factor models, performance attribution, and what a mean-variance optimiser would have done with the same eleven stocks

# Jensen's alpha: significant on every model we ran

Excess return regressed on risk factors. 72 monthly observations, Newey–West standard errors with four lags

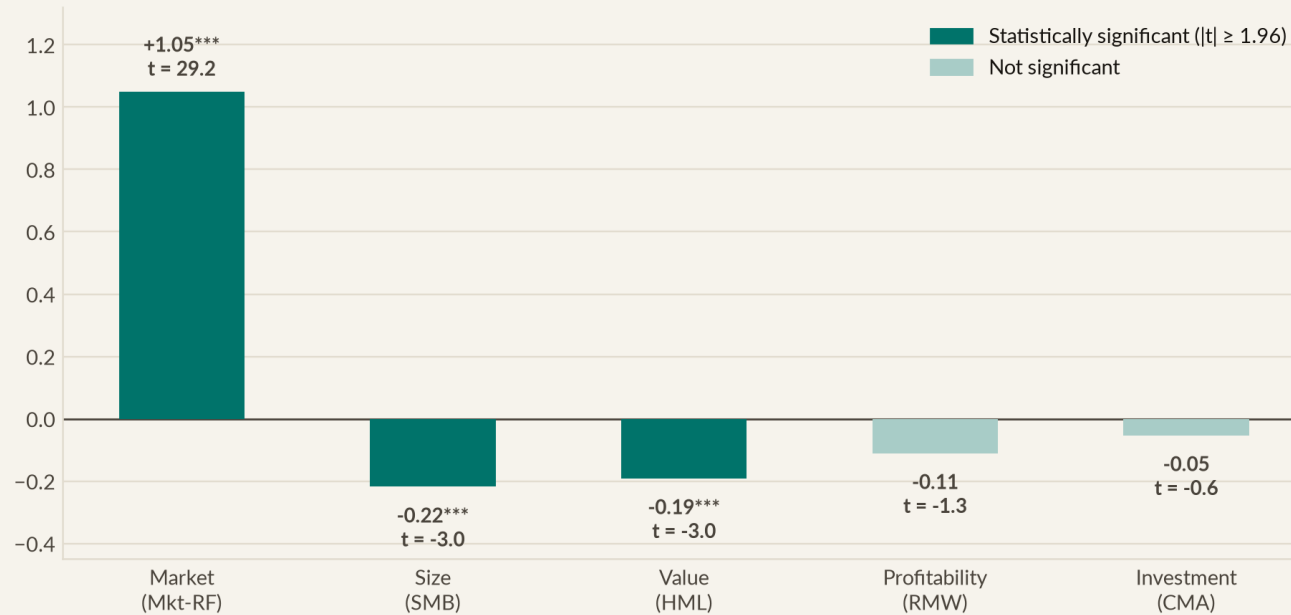
| MODEL                            | ANNUALISED ALPHA | ALPHA T-STAT | P-VALUE | MARKET BETA | ADJUSTED R <sup>2</sup> | INTERPRETATION                |
|----------------------------------|------------------|--------------|---------|-------------|-------------------------|-------------------------------|
| <b>CAPM (single factor)</b>      | 7.2%             | 2.19         | 0.029   | 1.005       | 0.872                   | Significant at the 5% level   |
| <b>Fama-French 3-factor</b>      | 5.9%             | 2.83         | 0.005   | 1.040       | 0.915                   | Significant at the 1% level   |
| <b>Fama-French 5-factor</b>      | 6.6%             | 3.06         | 0.002   | 1.050       | 0.914                   | Significant at the 1% level   |
| <b>FF5 + momentum (Carhart)</b>  | 6.5%             | 2.95         | 0.003   | 1.054       | 0.913                   | Significant at the 1% level   |
| <b>NASDAQ-100, same FF5 test</b> | 2.4%             | 1.41         | 0.159   | 1.099       | 0.943                   | Not statistically significant |

The test that matters. After controlling for the market, size, value, profitability, investment and momentum factors, the fund still delivered 6.6% a year that those factors cannot explain, with a t-statistic of 3.06 ( $p = 0.002$ ). The same test applied to the NASDAQ-100 returns an alpha of 2.4% that is statistically indistinguishable from zero. The benchmark's excess return is explained by its factor exposures. Ours is not.

## WAS IT SKILL?

# What the factor loadings say about us

A large-cap growth fund with market-like beta, which is the profile the mandate described



## READING THE COEFFICIENTS

**Market +1.05.** Essentially one-for-one with the market. We were not levered.

**Size -0.22.** Significantly negative: a large-cap fund, as intended.

**Value -0.19.** Significantly negative: a growth fund, as intended.

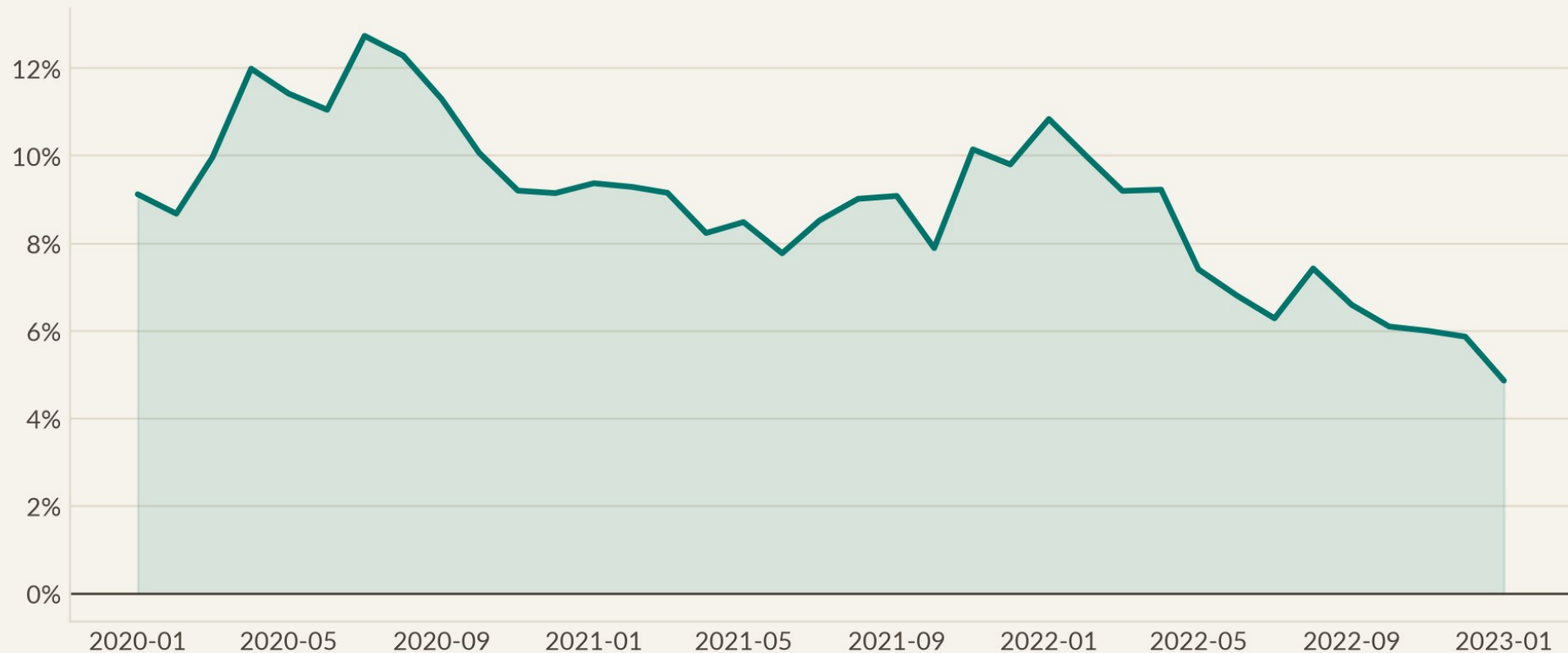
**Profitability -0.11, Investment -0.05.** Neither is significant, so no hidden quality tilt.

**Momentum +0.02.** Effectively zero. This was not a momentum-chasing strategy.

**Every loading sits where the January-2016 mandate said it would. There are no unexplained exposures hiding in this portfolio.**

# Alpha was persistent, not a single lucky year

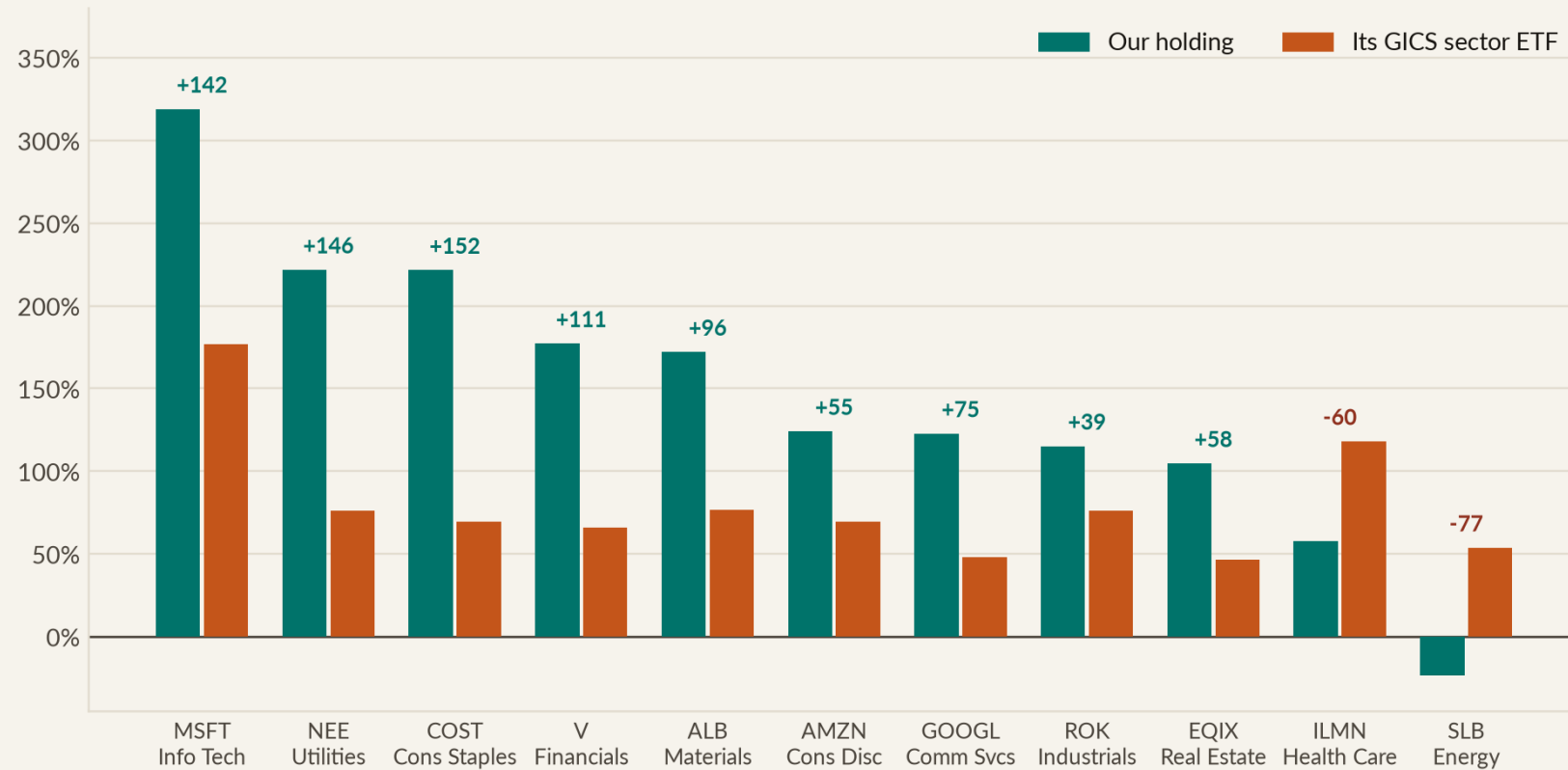
Rolling 36-month Jensen's alpha stays positive across every window in the sample



A single full-sample alpha can be produced by one exceptional year. This chart re-estimates the CAPM alpha on every rolling three-year window from Dec-2019 onward. It never turns negative — including the windows dominated by the 2022 drawdown, where the alpha comes from losing less rather than gaining more.

# Nine of eleven picks beat their own sector

Each holding measured against the GICS sector ETF it sits in, which is the fairest test of stock selection

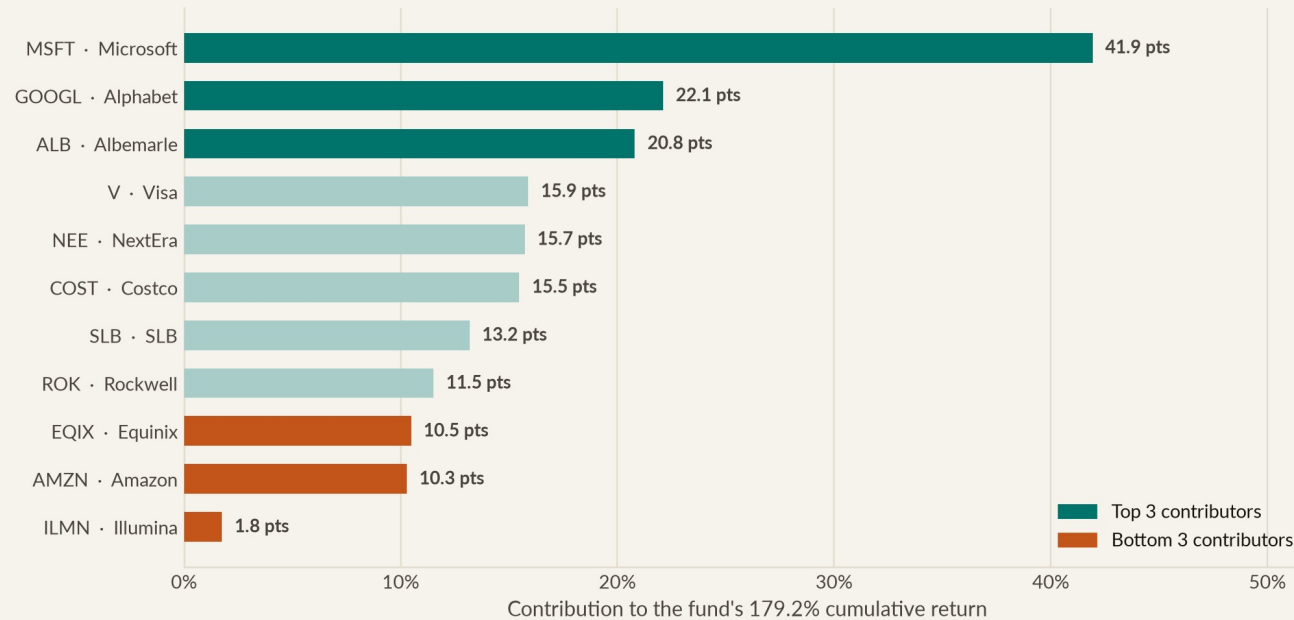


Only two picks lagged: Illumina, 60 points behind XLV, and Schlumberger, 77 points behind XLE. The standouts were Costco at 152 points ahead of XLP, NextEra 146 ahead of XLU and Microsoft 142 ahead of XLK. XLC did not exist until June 2018, so the Communication Services sleeve is spliced with XLK before that date.

## WAS IT SKILL?

# Where the 179% came from

Compounded contribution to cumulative return by holding. Microsoft alone delivered nearly a quarter of it



## TOP 3 AND BOTTOM 3

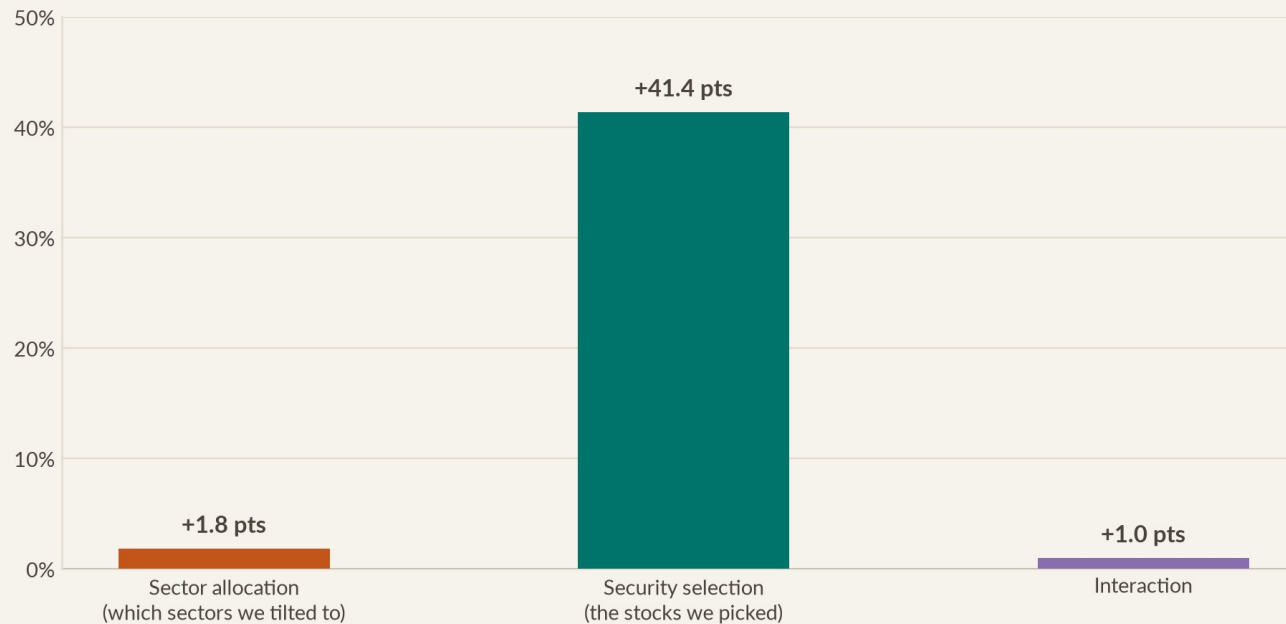
|              | CONTRIB. | TOTAL RETURN |
|--------------|----------|--------------|
| <b>MSFT</b>  | 41.9 pts | 319%         |
| <b>GOOGL</b> | 22.1 pts | 123%         |
| <b>ALB</b>   | 20.8 pts | 172%         |
| <b>EQIX</b>  | 10.5 pts | 105%         |
| <b>AMZN</b>  | 10.3 pts | 124%         |
| <b>ILMN</b>  | 1.8 pts  | 58%          |

Schlumberger is the instructive case. It lost 23.5% over six years, yet contributed a positive 13.2 points — because quarterly rebalancing kept buying it back to its 5% floor at successively lower prices, and it then returned +81% in 2022.

## WAS IT SKILL?

# 94% of the value-add came from stock picking

Brinson–Fachler attribution against an equal-weight benchmark of the eleven GICS sector ETFs



## DECOMPOSITION

**Allocation +1.8 pts.** The sector tilts, overweight technology and underweight energy, contributed very little.

**Selection +41.4 pts.** Almost all of the active return came from picking better stocks than the sector average.

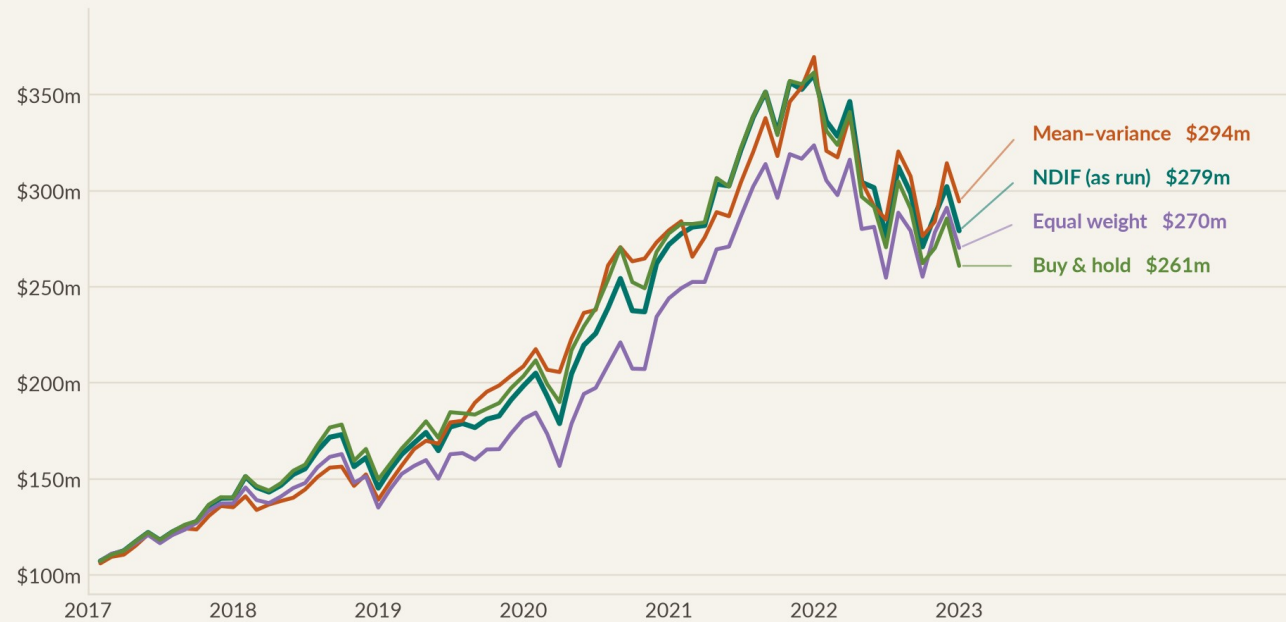
**Interaction +1.0 pt.** The residual from being overweight where we also picked well.

Why this matters for your decision: allocation skill is a macro call and is hard to repeat. Selection skill is a research process. That 94% of our value-add came from selection is the strongest argument that the result is repeatable.

## WAS IT SKILL?

# Would an optimiser have done better?

A mean-variance model given three years of pre-launch data would have added 1.1% a year, by taking concentration risk we were not permitted to take



| CONSTRUCTION RULE              | CAGR  | SHARPE |
|--------------------------------|-------|--------|
| NDIF as run                    | 18.7% | 0.94   |
| Mean-variance (2014-16 inputs) | 19.7% | 1.07   |
| Equal weight (1/N)             | 18.0% | 0.90   |
| Buy & hold                     | 17.3% | 0.88   |
| Perfect-foresight optimum      | 22.8% | 1.32   |

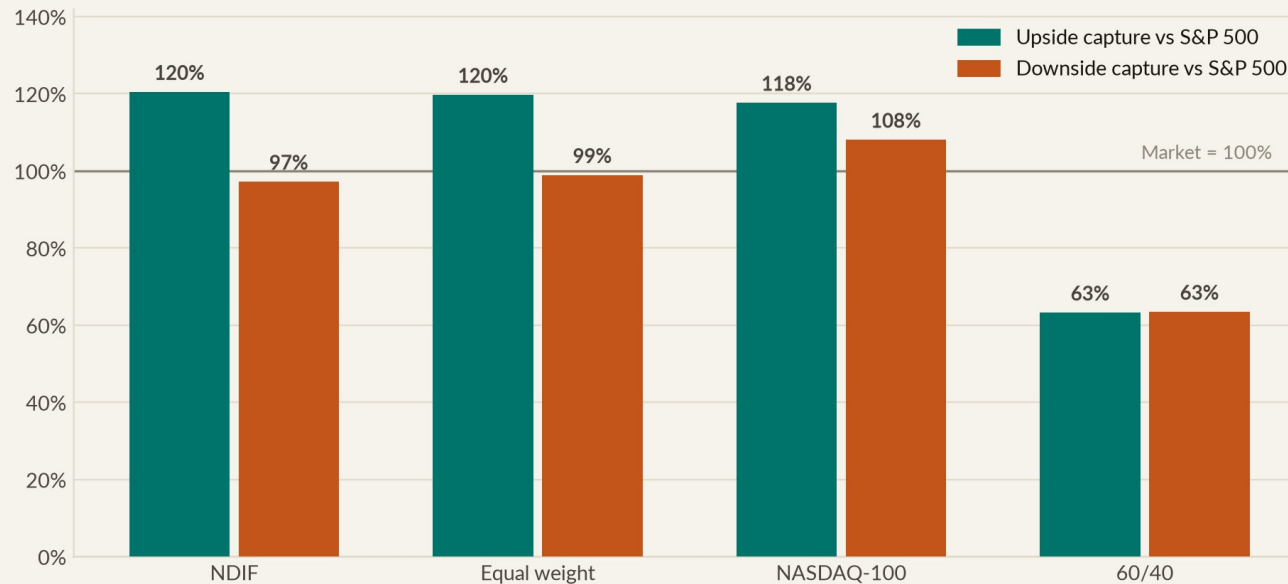
The optimiser's solution put 25% in Equinix and 25% in NextEra — breaching our 16% position cap — and zero in Alphabet, Visa, Albemarle and Schlumberger, breaching the 5% sector floor four times over. It bought a better backtest by discarding the mandate.

Even with perfect foresight, the best achievable was 22.9% a year. We captured 82% of a result nobody could have known in advance.

## WAS IT SKILL?

# 120% of the upside, 97% of the downside

The asymmetry that produced the result: most of the market's gains, without most of its losses



## THE ASYMMETRY

**Upside capture 120%.** In months the S&P 500 rose, we rose 20% more than it did.

**Downside capture 97%.** In months it fell, we fell slightly less.

**The NASDAQ-100, by contrast,** captured 118% of the upside but 108% of the downside: more of the gains, and more of the losses.

**50 positive months of 72.** A 69.4% hit rate against 65.3% for the NASDAQ-100.

**Those two numbers, 120 up and 97 down, are the compact statement of what the fund did. Everything else in this deck elaborates on them.**

# 06

---

## The verdict

Marking the fund against the objective we set in 2016, the fees you paid, and our recommendation on what happens next

## THE VERDICT

# Marking ourselves against the 2016 objective

Every commitment we made when we asked you for the mandate, scored against what we actually delivered

| WHAT WE COMMITTED TO IN JANUARY 2016                   | TARGET              | DELIVERED                                                                 | VERDICT  |
|--------------------------------------------------------|---------------------|---------------------------------------------------------------------------|----------|
| <b>Net annualised return</b>                           | 8-10% p.a.          | 17.5% p.a. net of all fees                                                | Exceeded |
| <b>Cumulative net return over the fund life</b>        | 55-70%              | 163% net                                                                  | Exceeded |
| <b>Outperform the S&amp;P 500</b>                      | ~2% p.a. net        | +6.2% p.a. net                                                            | Exceeded |
| <b>Outperform the NASDAQ-100 (primary benchmark)</b>   | Beat it             | +3.4% p.a. gross                                                          | Met      |
| <b>Maintain all eleven GICS sectors, 5% floor</b>      | No sector abandoned | Held throughout, all 24 quarters                                          | Met      |
| <b>Cap the largest position at 16%</b>                 | 16% cap             | Held at every rebalance; MSFT drifted to a peak of 17.6% within a quarter | Met      |
| <b>Rebalance quarterly with a 5-point drift rule</b>   | 24 rebalances       | 24 executed; drift never exceeded 2.9 pts, so the rule never fired        | Met      |
| <b>Deliver a mid-life strategic review at end-2019</b> | Formal review       | Conducted; target weights reconfirmed                                     | Met      |
| <b>Keep the drawdown below the theme benchmark's</b>   | Beat NASDAQ-100     | -24.9% vs -32.6%                                                          | Met      |
| <b>Generate return the market cannot explain</b>       | Positive alpha      | +6.6% p.a., $t = 3.06$ , $p = 0.002$                                      | Exceeded |

THE VERDICT

# Gross to net: what you actually kept

0.90% management fee plus 10% of the excess over the NASDAQ-100 hurdle, subject to the high-water mark



|                  | P.A.   | OVER SIX YEARS |
|------------------|--------|----------------|
| Gross return     | 18.7%  | 179%           |
| Management fee   | −0.90% | −5.3%          |
| Performance fee  | −0.25% | −1.5%          |
| Net to investors | 17.51% | 163%           |

Total fee load was 1.15% a year — 0.90% base plus 0.25% of performance fee, because the hurdle is the NASDAQ-100 rather than zero and we only cleared it by 2.5 points a year.

A passive NASDAQ-100 ETF at 0.20% would have cost \$5.7m less over the six years and returned \$235m. We returned \$279m, so investors kept \$38.7m after paying for active management.

## THE VERDICT

# Our recommendation: the fund should continue

With three changes to the mandate, which we would want your agreement on before raising the successor vehicle

---

## THE CASE FOR CONTINUING

**The alpha is real and it is repeatable.** 6.6% a year that five risk factors cannot explain, with 94% of it from security selection rather than sector timing.

**The thesis is not exhausted.** Cloud is still a minority of enterprise IT spend; electrification and genomics are earlier still. The 2022 de-rating repriced the multiple, not the demand.

**The construction discipline was tested and held.** Sector floors, the position cap and quarterly rebalancing were all worth measurable money — most visibly in 2022.

---

## WHAT WE WOULD CHANGE

**Widen the tail-scenario assumptions.** Our Scenario C assumed a 4% loss. The real one cost 22.5%. The framework identified the regime correctly and sized it badly.

**Add a valuation discipline to the entry rule.** Illumina, our worst selection decision, was a correct thesis bought without a price constraint.

**Review the position cap upward, carefully.** The 16% cap forced us to trim Microsoft, our single best decision, at 19 of the 24 rebalances. We would seek 20%, with the sector floors untouched.

NORTHPOINT

# Thank you.

---

I am happy to take questions on any part of this review.

An appendix follows with the full methodology, the complete metric tables, the per-holding statistics, and the full factor-model output.

Callum O'Connor · Portfolio Manager · Northpoint Capital Partners  
FINC13-303 Portfolio Analysis and Investments · Student ID 14053836



---

# Appendix

Methodology and data sources, full metric tables, per-holding statistics, complete factor-model output, and supporting exhibits

## APPENDIX

# A1 · Methodology, data and assumptions

Everything in this deck is reproducible from the accompanying Python analysis and Excel workbook

| ITEM                       | TREATMENT                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Return data                | Monthly total returns from dividend- and split-adjusted closing prices, Jan 2017 – Dec 2022 (72 observations per series)               |
| Portfolio construction     | Target weights per the January-2016 mandate; drift within the quarter, restored to target at each calendar quarter-end (24 rebalances) |
| Primary benchmark          | NASDAQ-100 total return, proxied by the QQQ ETF so that dividends are included (the ^NDX index itself is price-only)                   |
| Secondary benchmark        | S&P 500 total return, proxied by the SPY ETF on the same basis                                                                         |
| Sector benchmarks          | The eleven SPDR GICS sector ETFs. XLC launched Jun-2018, so XLK is used for the Communication Services sleeve before that date         |
| Risk-free rate             | Fama-French one-month Treasury bill series; mean 1.25% p.a. over the sample                                                            |
| Risk factors               | Kenneth R. French Data Library: Mkt-RF, SMB, HML, RMW, CMA and the momentum factor, monthly, US research series                        |
| Regressions                | OLS with Newey-West (HAC) standard errors, four lags, to correct for autocorrelation and heteroskedasticity in monthly returns         |
| Annualisation              | Returns compounded geometrically; volatility scaled by $\sqrt{12}$ ; ratios computed on monthly data then annualised                   |
| Attribution                | Brinson-Fachler, summed arithmetically across 72 months against an equal-weight benchmark of the eleven sector ETFs                    |
| Mean-variance optimisation | SLSQP, long-only, maximum Sharpe, 25% position bound; in-sample window Jan 2014 – Dec 2016 for the ex-ante case                        |
| Excluded                   | Transaction costs, market impact, taxes and cash drag. All returns are gross of these unless a slide states otherwise                  |

## APPENDIX

## A2 · Full performance and risk metrics

All strategies and benchmarks, 72 monthly observations, January 2017 – December 2022

| METRIC                               | NDIF   | EQUAL WEIGHT | MEAN-VARIANCE | NASDAQ-100 | S&P 500 | 60/40  |
|--------------------------------------|--------|--------------|---------------|------------|---------|--------|
| <b>Cumulative return</b>             | 179%   | 170%         | 194%          | 135%       | 90%     | 52%    |
| <b>Annualised return</b>             | 18.7%  | 18.0%        | 19.7%         | 15.3%      | 11.3%   | 7.2%   |
| <b>Annualised volatility</b>         | 19.1%  | 19.4%        | 17.4%         | 20.2%      | 17.1%   | 11.1%  |
| <b>Mean monthly return</b>           | 1.59%  | 1.54%        | 1.64%         | 1.36%      | 1.02%   | 0.63%  |
| <b>Median monthly return</b>         | 2.73%  | 2.26%        | 2.69%         | 2.20%      | 1.83%   | 1.13%  |
| <b>Skewness</b>                      | -0.47  | -0.38        | -0.74         | -0.39      | -0.48   | -0.50  |
| <b>Excess kurtosis</b>               | 0.07   | 0.10         | 0.62          | -0.04      | 0.35    | 0.56   |
| <b>Best month</b>                    | 14.5%  | 14.0%        | 12.4%         | 15.0%      | 12.7%   | 8.3%   |
| <b>Worst month</b>                   | -12.1% | -11.4%       | -13.2%        | -13.6%     | -12.5%  | -7.7%  |
| <b>Sharpe ratio</b>                  | 0.94   | 0.90         | 1.07          | 0.75       | 0.65    | 0.59   |
| <b>Sortino ratio</b>                 | 1.51   | 1.45         | 1.69          | 1.18       | 0.98    | 0.88   |
| <b>Treynor ratio</b>                 | 0.173  | 0.163        | 0.238         | 0.139      | 0.111   | 0.102  |
| <b>Calmar ratio</b>                  | 0.75   | 0.85         | 0.78          | 0.47       | 0.47    | 0.36   |
| <b>Maximum drawdown</b>              | -24.9% | -21.3%       | -25.2%        | -32.6%     | -23.9%  | -20.0% |
| <b>Downside deviation</b>            | 11.9%  | 12.0%        | 11.0%         | 12.9%      | 11.3%   | 7.4%   |
| <b>VaR 95% (monthly)</b>             | -8.8%  | -9.3%        | -7.5%         | -8.8%      | -8.5%   | -5.0%  |
| <b>CVaR 95% (monthly)</b>            | -10.2% | -10.3%       | -10.5%        | -10.5%     | -9.8%   | -6.8%  |
| <b>Beta vs S&amp;P 500</b>           | 1.04   | 1.07         | 0.78          | 1.10       | 1.00    | 0.64   |
| <b>Tracking error vs S&amp;P 500</b> | 7.0%   | 6.6%         | 11.8%         | 7.6%       | 0.0%    | 6.4%   |
| <b>Upside capture</b>                | 120%   | 120%         | 101%          | 118%       | 100%    | 63%    |
| <b>Downside capture</b>              | 97%    | 99%          | 63%           | 108%       | 100%    | 63%    |
| <b>Terminal value on \$100m</b>      | \$279m | \$270m       | \$294m        | \$235m     | \$190m  | \$152m |

## APPENDIX

## A3 · Per-holding statistics

Every position measured over the full six years on the same basis as the portfolio

| TICKER       | GICS SECTOR  | WEIGHT | TOTAL RETURN | CAGR  | VOLATILITY | SHARPE | MAX DD | BETA | CAPM ALPHA | T-STAT | CONTRIBUTION |
|--------------|--------------|--------|--------------|-------|------------|--------|--------|------|------------|--------|--------------|
| <b>MSFT</b>  | Info Tech    | 16%    | 319%         | 27.0% | 20.4%      | 1.23   | -30.5% | 0.93 | 16.4%      | 2.76   | 41.9 pts     |
| <b>GOOGL</b> | Comm Svcs    | 14%    | 123%         | 14.3% | 24.1%      | 0.63   | -40.4% | 1.09 | 3.8%       | 0.70   | 22.1 pts     |
| <b>AMZN</b>  | Cons Disc    | 12%    | 124%         | 14.4% | 32.2%      | 0.54   | -52.1% | 1.23 | 4.4%       | 0.37   | 10.3 pts     |
| <b>V</b>     | Financials   | 10%    | 177%         | 18.5% | 22.2%      | 0.83   | -27.3% | 0.96 | 8.8%       | 1.59   | 15.9 pts     |
| <b>EQIX</b>  | Real Estate  | 9%     | 105%         | 12.7% | 22.7%      | 0.59   | -32.1% | 0.59 | 7.8%       | 0.87   | 10.5 pts     |
| <b>ILMN</b>  | Health Care  | 8%     | 58%          | 7.9%  | 36.3%      | 0.36   | -62.8% | 1.14 | 0.6%       | 0.05   | 1.8 pts      |
| <b>ROK</b>   | Industrials  | 7%     | 115%         | 13.6% | 31.4%      | 0.53   | -42.3% | 1.44 | 1.3%       | 0.20   | 11.5 pts     |
| <b>NEE</b>   | Utilities    | 7%     | 222%         | 21.5% | 19.6%      | 1.04   | -23.5% | 0.50 | 16.5%      | 3.14   | 15.7 pts     |
| <b>COST</b>  | Cons Staples | 6%     | 222%         | 21.5% | 22.0%      | 0.95   | -20.3% | 0.79 | 13.7%      | 1.95   | 15.5 pts     |
| <b>ALB</b>   | Materials    | 6%     | 172%         | 18.1% | 43.8%      | 0.57   | -58.3% | 1.47 | 9.3%       | 0.63   | 20.8 pts     |
| <b>SLB</b>   | Energy       | 5%     | -24%         | -4.4% | 49.0%      | 0.14   | -81.6% | 1.78 | -11.6%     | -0.64  | 13.2 pts     |
| <b>NDIF</b>  | All eleven   | 100%   | 179%         | 18.7% | 19.1%      | 0.94   | -24.9% | 1.04 | 7.2%       | 2.19   | 179.2 pts    |

Only Microsoft (t = 2.76) and NextEra (t = 3.14) clear the 1.96 threshold on their own; Costco misses it by a whisker at 1.95. The portfolio's alpha is more significant than any constituent's, which is the diversification benefit doing real statistical work.

## APPENDIX

## A4 · Complete factor-model output

Coefficient, t-statistic and significance for every factor across all four models

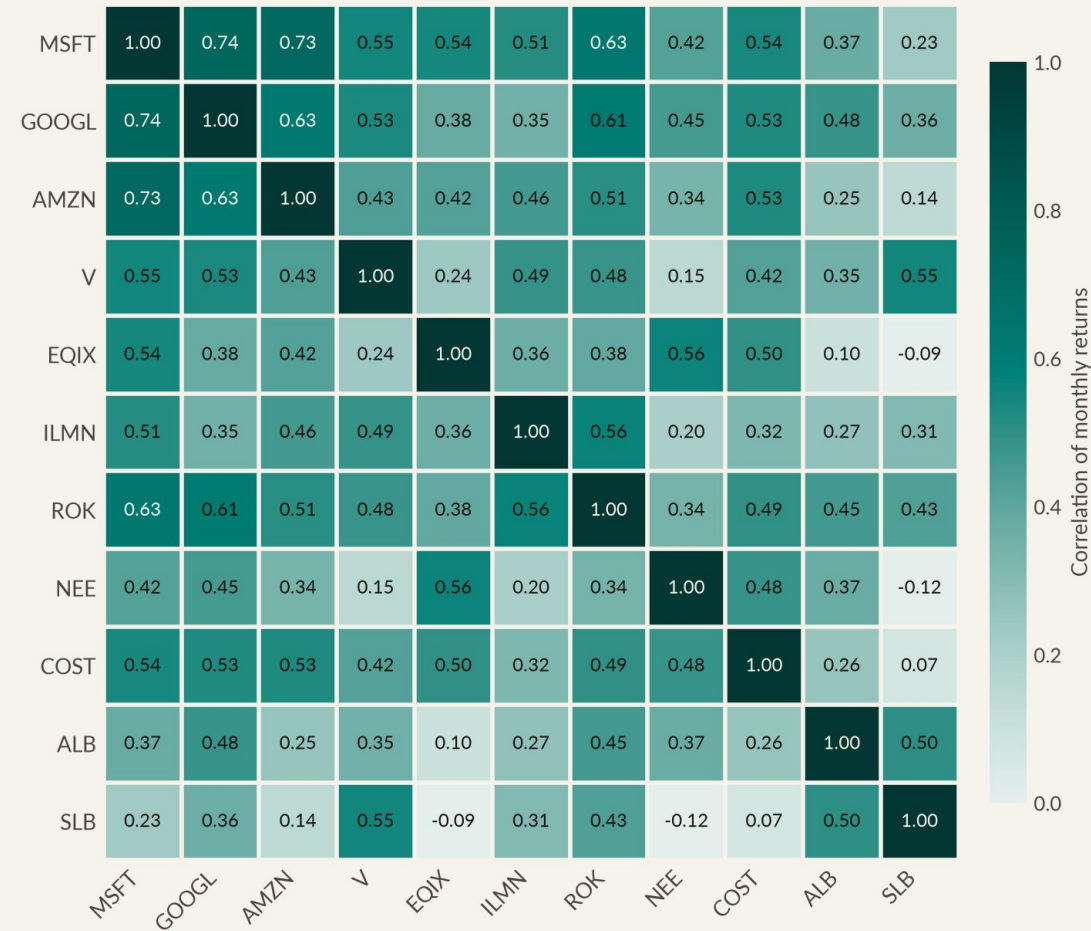
| FACTOR                        | CAPM                     | FAMA-FRENCH 3            | FAMA-FRENCH 5            | FF5 + MOMENTUM           |
|-------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| <b>Alpha (annualised)</b>     | 7.2%**<br>(t = 2.19)     | 5.9%***<br>(t = 2.83)    | 6.6%***<br>(t = 3.06)    | 6.5%***<br>(t = 2.95)    |
| <b>Market (Mkt-RF)</b>        | +1.005***<br>(t = 15.75) | +1.040***<br>(t = 25.66) | +1.050***<br>(t = 29.16) | +1.054***<br>(t = 25.88) |
| <b>Size (SMB)</b>             | —                        | -0.164**<br>(t = -2.30)  | -0.218***<br>(t = -3.01) | -0.209**<br>(t = -2.50)  |
| <b>Value (HML)</b>            | —                        | -0.236***<br>(t = -5.61) | -0.191***<br>(t = -3.05) | -0.184***<br>(t = -2.76) |
| <b>Profitability (RMW)</b>    | —                        | —                        | -0.111<br>(t = -1.30)    | -0.104<br>(t = -1.23)    |
| <b>Investment (CMA)</b>       | —                        | —                        | -0.053<br>(t = -0.64)    | -0.060<br>(t = -0.72)    |
| <b>Momentum (MOM)</b>         | —                        | —                        | —                        | +0.022<br>(t = 0.29)     |
| <b>Adjusted R<sup>2</sup></b> | 0.872                    | 0.915                    | 0.914                    | 0.913                    |
| <b>Observations</b>           | 72                       | 72                       | 72                       | 72                       |

\*\*\* significant at 1% \*\* significant at 5% \* significant at 10%. Newey-West heteroskedasticity- and autocorrelation-consistent standard errors, four lags.

Dependent variable is the fund's monthly return in excess of the one-month Treasury bill rate.

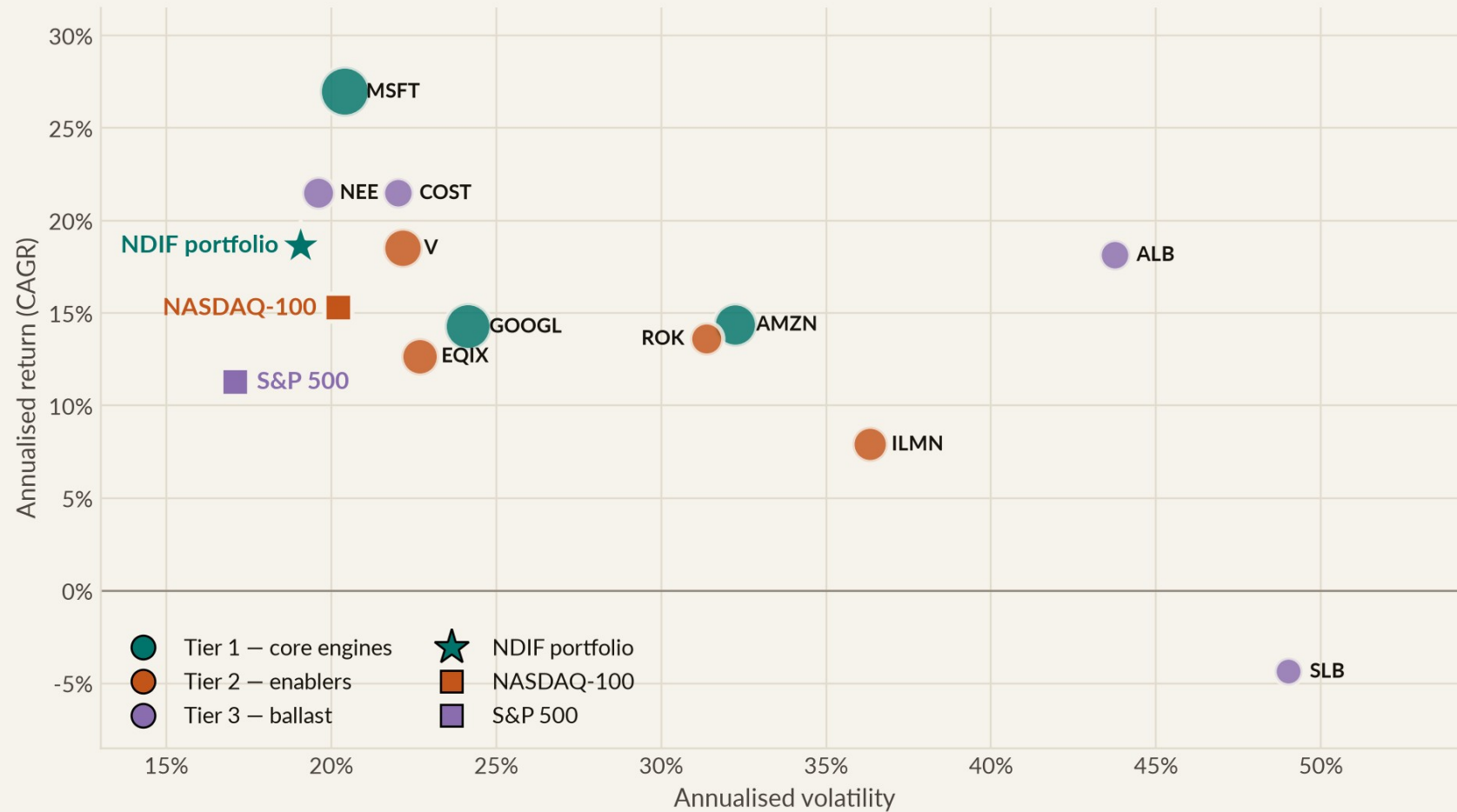
## A5 · Correlation structure of the holdings

Average pairwise correlation of 0.40, which is real diversification inside a single theme



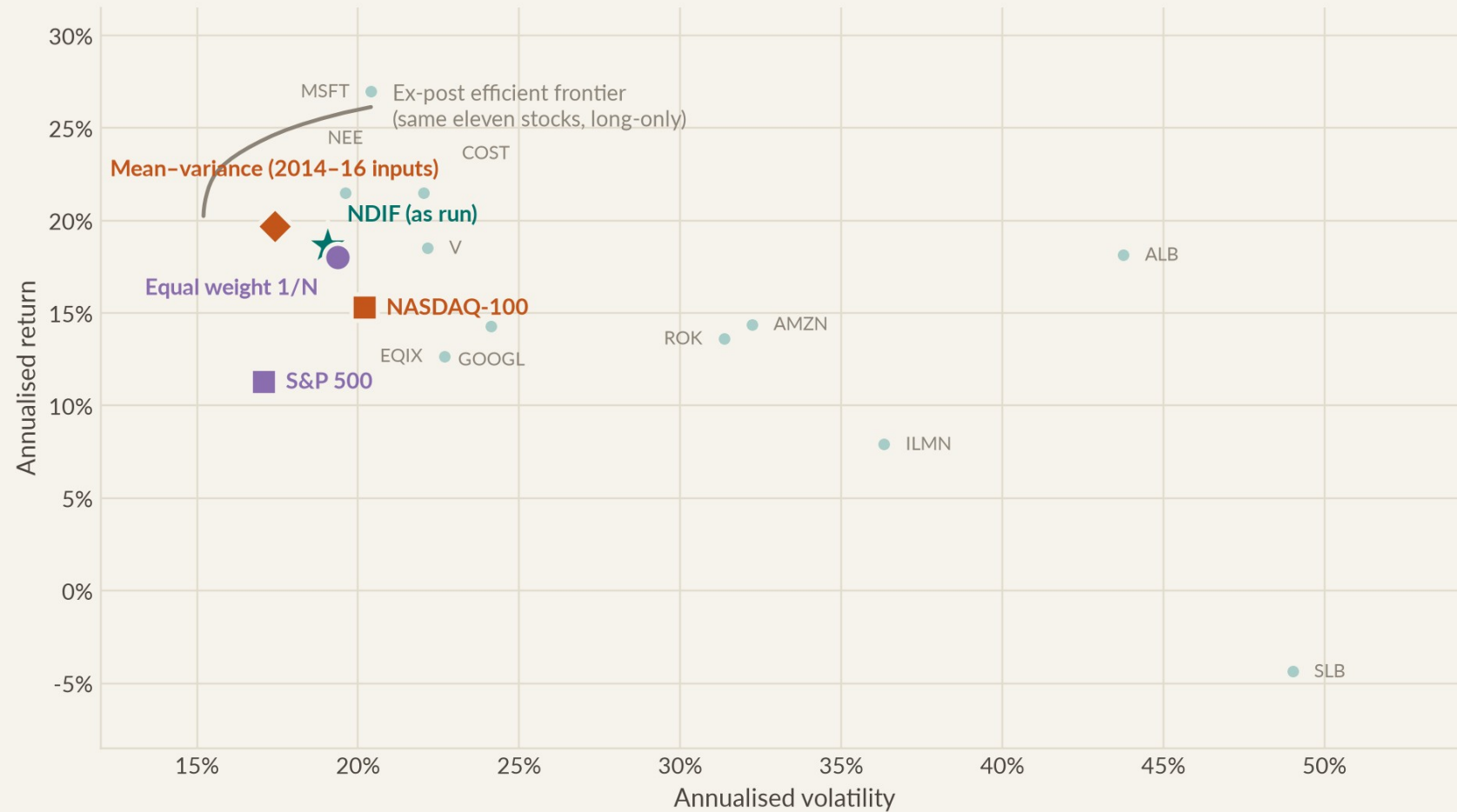
## A6 · The risk–return map

Every holding, the portfolio and both benchmarks on realised risk and return



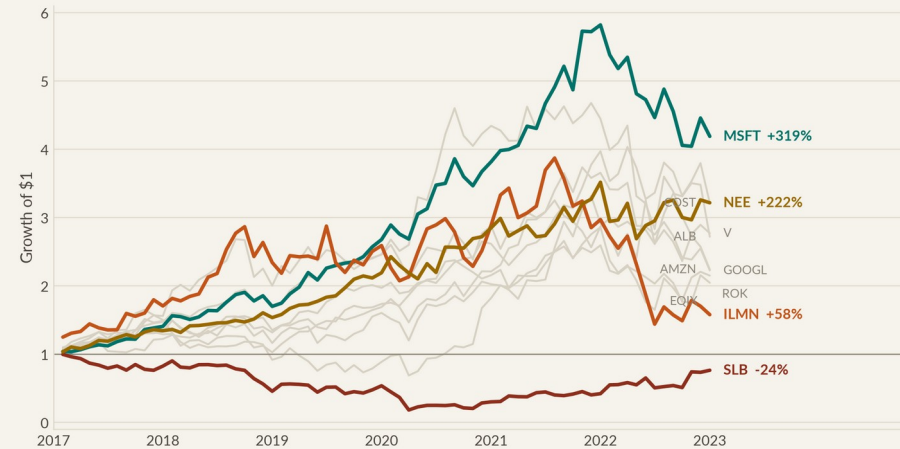
## A7 · The efficient frontier of our own holdings

Where the fund actually landed against the best that was achievable ex post



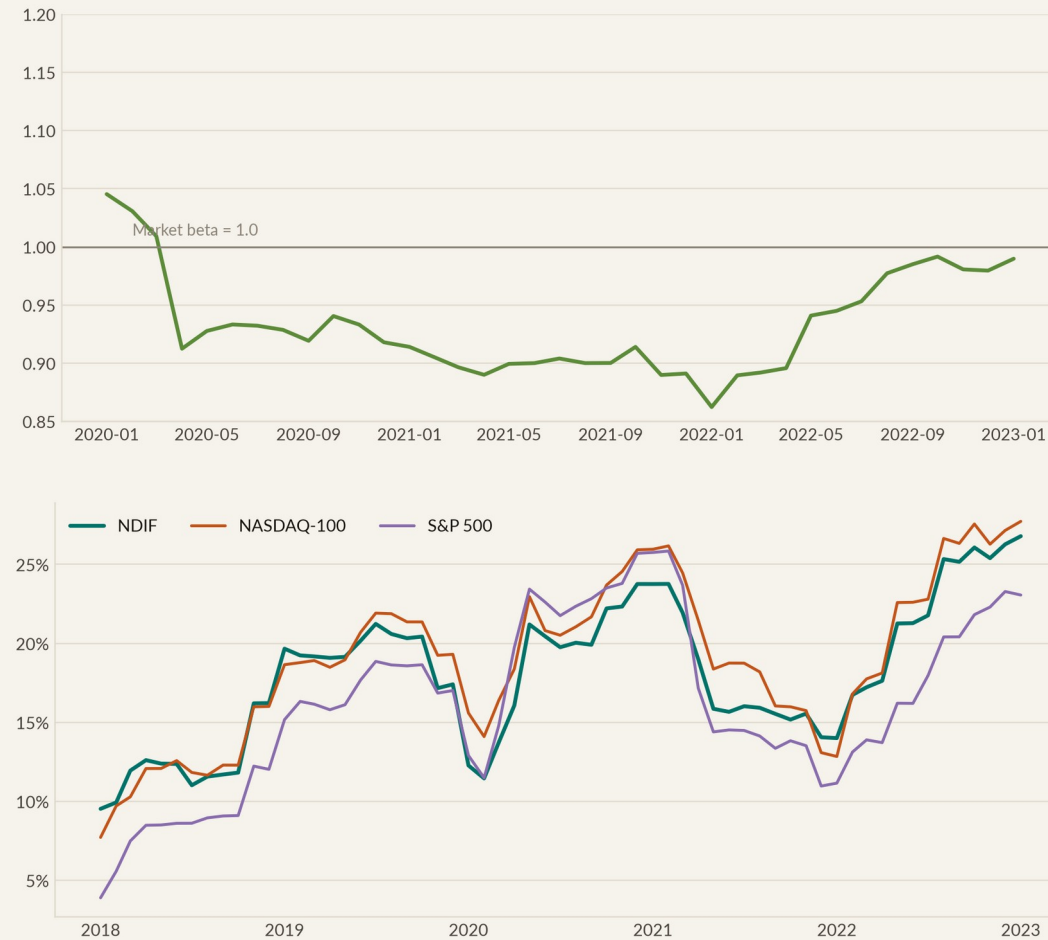
## A8 · Distribution comparison and dispersion

How the fund's return distribution compares to its benchmarks, and the spread across individual holdings



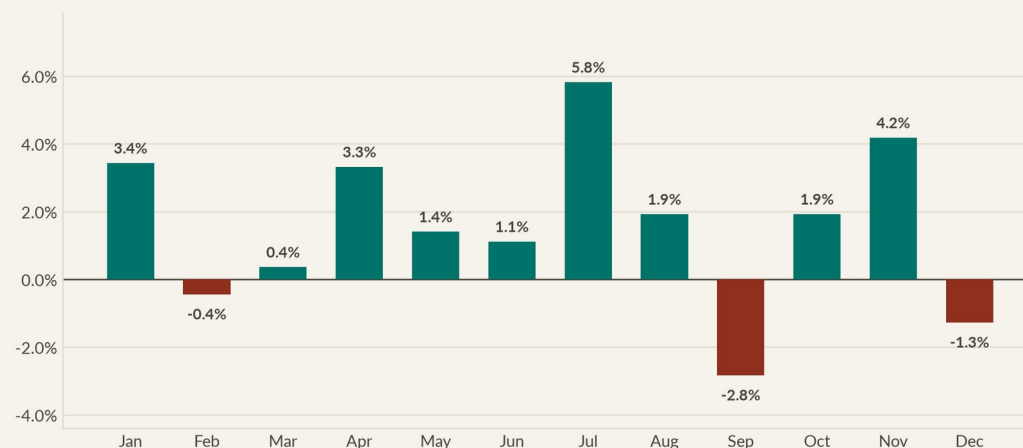
# A9 · Rolling beta and volatility

Stability of the fund's market exposure and total risk through the sample



# A10 · Month-of-year seasonality and the fee comparison

Supporting detail on the seasonality discussion and the fee benchmarking



| COMPARATOR                         | FEE P.A.     | 6-YEAR FEE ON \$100M | RETURN DELIVERED | TERMINAL VALUE |
|------------------------------------|--------------|----------------------|------------------|----------------|
| Passive S&P 500 ETF (SPY)          | 0.0945%      | \$0.57m              | 90%              | \$190m         |
| Passive NASDAQ-100 ETF (QQQ)       | 0.20%        | \$1.20m              | 135%             | \$235m         |
| Typical active equity fund         | 1.25%        | \$7.50m              | —                | —              |
| Northpoint Digital Innovation Fund | 1.15% all-in | \$6.90m              | 179%             | \$279m         |

Fees are calculated on a static \$100m for comparability. Northpoint's all-in load of 1.15% sits 0.95 points above a passive NASDAQ-100 ETF, which costs \$5.7m more over six years. The fund delivered \$44.4m of additional terminal value, so investors kept \$38.7m after paying for active management.

## APPENDIX

## A11 · References and declarations

Sources, academic references, and the required statement on the use of generative AI

## REFERENCES

Brinson, G. P., Hood, L. R., & Beebower, G. L. (1986). Determinants of portfolio performance. *Financial Analysts Journal*, 42(4), 39–44.

DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal versus naive diversification. *The Review of Financial Studies*, 22(5), 1915–1953.

Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.

Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.

Jensen, M. C. (1968). The performance of mutual funds in the period 1945–1964. *The Journal of Finance*, 23(2), 389–416.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91.

Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

Sharpe, W. F. (1966). Mutual fund performance. *The Journal of Business*, 39(1), 119–138.

Sortino, F. A., & Price, L. N. (1994). Performance measurement in a downside risk framework. *The Journal of Investing*, 3(3), 59–64.

Treynor, J. L. (1965). How to rate management of investment funds. *Harvard Business Review*, 43(1), 63–75.

Data: Kenneth R. French Data Library (factor and risk-free series); dividend-adjusted monthly closing prices for all eleven holdings, the eleven SPDR sector ETFs, QQQ, SPY and AGG.

## GENERATIVE AI USE DECLARATION

**Assessment classification.** AI-Supported. Generative AI was permitted for the preparation of the analysis and slides; the presentation itself is delivered by the author.

**Tools used.** Generative AI assistant for the Python analysis code, chart generation and slide drafting; Grammarly for language editing.

**Purpose.** Writing the return, risk and regression code; producing the visualisations; structuring the deck; drafting speaker notes.

**Human refinement.** As Portfolio Manager I set the analytical scope, chose the benchmarks and factor models, interpreted every result, wrote the investment narrative and recommendation, and verified the outputs against the underlying data.

**Data integrity.** All market data was retrieved programmatically from primary sources. No figure in this deck was generated by an AI model without a reproducible calculation behind it.